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 SURVEY

A Survey on Hardware Security: Current Trends and Challenges

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ABSTRACT Hardware security has become a critical concern due to the globalization of the Integrated Circuit (IC) supply chain and the complex network connections of computing-intensive devices. Hardware security is essential in the modern world, as more and more connectivity of the Internet of Things (IoT) has paved the way for improvements in personalized healthcare, communication between home (or office) equipment, and the promise of self-driving cars, airplanes, and smart grid systems. As a result, recently, this field has attracted particular attention from the researcher. A significant amount of research has already been conducted to detect, defend, and create resiliency against hardware attacks. This paper investigates different hardware security approaches in the state-of-the-art literature and outlines their advantages and drawbacks. This paper also provides a comprehensive study of hardware security attacks, and their features, and categorizes different attack types. Finally, it presents the current challenges of the hardware security field and the impact of each one on the IC production supply chain.

INDEX TERMS Machine learning, the Internet of Things (IoT), AES, hardware security, privacy, neural networks.

I. INTRODUCTION

Hardware security is a broad term that describes the use of each hardware component to guarantee the security properties to all stages of the IC supply chain, starting from the silicon semiconductor collection, IC design, specification, outsourcing IPs/ICs, to the after-life cycle of an IC. These security properties can be applied to both information processed by the hardware design and the hardware implementation itself. There are six security properties in the literature: Dependability, Confidentiality, Integrity, Isolation, Constant time, and Quantitative Security Properties [1]. These security characteristics could be applied to both the hardware implementation and the data in the hardware design processes. Therefore, there are two objectives for the secure hardware design. The first one is to develop a safe hardware design at the targeted performance level. The second one is

that the designer must ensure that the security features of the design can be enforced against an experienced opponent who monitors or tampers the secure hardware design. The idea behind hardware-assisted security is to use hardware to enable trustworthy computing by giving the software root of trust [2].

Hardware security is essential in the modern world, as more and more connectivity of the Internet of Things (IoT) has paved the way for improvements in personalized healthcare, communication between home (or office) equipment, and the promise of self-driving cars, airplanes, and smart grid systems. Moreover, real-time applications, such as smart factories, delivery drones, search-and-rescue drones are gaining popularity. For these real-time systems to function correctly, specific deadlines must be met, and security needs to be ensured. Even though cryptographic algorithms are crucial in preventing key leakage, attackers continue to exploit hardware implementation defects to gain access to private data [3], which indicates the need for hardware security.

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Providing hardware security can strengthen the overall system security by providing an extra layer of security with the software security.

An Integrated Circuit (IC) comprises many Intellectual Property (IP) cores, such as a CPU, memory, network-on-chip, controllers, converters, input/output devices, etc. As the ICs are scaling down, their cost and complexity are also increasing. As a result, IC production is shifted overseas to reduce cost and complexity, making the IC production supply chain more prone to hardware attacks. This diversity of the IC supply chain creates more scope for attackers to insert malicious circuitry or codes in the design. A typical integrated circuit supply chain is shown in Fig. 1. Third-party vendors, after-life cycle, and market are painted blue to differentiate the outside party involvement in the IC supply chain. The brown part in the figure shows the in-house IC production flow.

Moreover, critical control and communication functions assume that the hardware platform they are implemented on is secure, but this assumption is not always valid. For example, [4] reports that backdoors could be used to rheostat weapon control systems, transportation systems, and nuclear power plants. Counterfeit electronics, a security vulnerability in hotel keycards, and a security vulnerability in parking payment cards are examples of practical hardware attacks [5].

In addition, open-source tools of modern computing systems, for example, commercial FPGA CAD tools and open-source FPGA CAD tools, have opened new doors for remote attacks without requiring physical access to the victim. Therefore, computing hardware is getting more prone to various emerging security threats. However, unintentional design flaws [3], system side effects [6], and malicious design alterations [7] are possible causes of hardware vulnerability. They frequently aim to steal intellectual property, secure systems, machine learning (ML) models, and cryptographic processes.

Hardware security is beneficial in the Internet of Things (IoT) environment. The IoT defines as the connection between physical objects over the Internet via smart computing devices [8], [9], [10]. It has expanded its arm towards every aspect of modern life, including the critical sector in applications like e-health care, banking, energy sector, defense applications, etc. Nowadays, all types of businesses, hospitals, companies, shopping malls, financial institutions, home appliances, and space science research facilities are gradually converting to smart platforms, connecting all electronic devices for collecting, processing, and transmitting data quickly. Moreover, at a CAGR of 26.4% over the forecast period, the worldwide IoT market is expected to increase from 478.36 billion dollars in 2022 to 2,465.26 billion dollars in 2029 [11]. However, this tremendous dependence on IoT devices creates many security holes at both software and hardware levels. Leaking information out of IoT nodes is not very difficult due to its open structure and participation of third parties [12], [13], [14], [15]. The IoT system is under attack not only at the software level but also at the

hardware level. Since these IoT devices can occasionally be installed remotely, some foreign objects or a leakage channel can be created by authorized individuals at the IoT system hardware level [16]. Enterprises can intentionally or unintentionally introduce hardware security vulnerabilities in IoTs during device design, manufacturing, integration, and distribution [17]. Hardware vulnerability in the IoT sector includes IP design modification, insertion of malicious contents in the form of hardware Trojan, Side-channel leakage, insecure tamper mechanism, PCB design modification, etc. Few methods are available to test vulnerability detection in pre-silicon and post-silicon stages, but they provide specific levels of effectiveness. These test methodologies, however, are used to evaluate a particular security vulnerability and do not take a systematic approach to tackle comprehensive hardware security testing of IoT devices. Additionally, few laboratories offer testing services because testing hardware vulnerability is too expensive and time-consuming. Low-cost IoT device manufacturers in medium- and small-sized sectors moreover, do not have enough resources to begin hardware testing.

As the flexible & adaptable characteristics of hardware-based data security systems make them an excellent fit for IoT in all its applications [18]. So, it is essential to identify potential security threats in IoT devices before deploying them into the field and understand how the various testing methods can be applied systemically to detect device-level vulnerabilities [19].

Security professionals have been working extremely hard to provide defense strategies and propose frameworks to identify hardware vulnerabilities and strengthen hardware design against a variety of known hardware and emerging attacks. However, balancing security over resource usage is still a challenging issue. In this work, several methods and frameworks proposed in the literature are presented, showing the efficacy and drawbacks of the proposed methods for specific attacks to help practitioners to choose a method/framework for that particular hardware attack. This research also accumulated several hardware security challenges so that more research can be done in that area.

The contributions of this paper are summarized in the following points:

- A review of the type of security attacks that are considered in any design including hardware and software attacks.
- A list of the unique challenges toward the existing hardware security methods.
- A comprehensive review of hardware security approaches and current trends.
- A tabular comparison between the current hardware security methods.
- A comprehensive review of hardware attack approaches and current trends.
- A tabular comparison between hardware attack approaches.

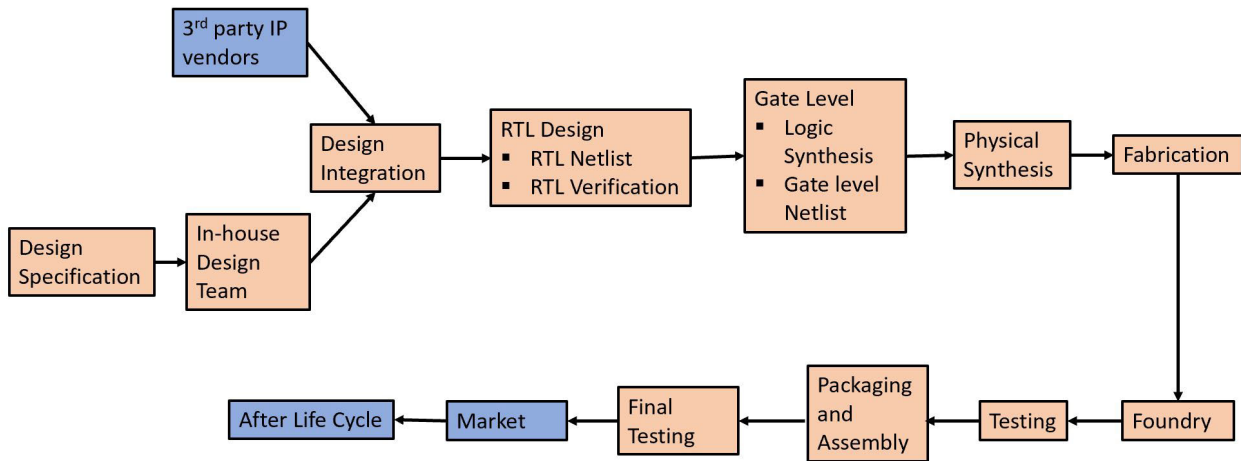


FIGURE 1. Integrated circuit (IC) design supply chain.

The remainder of this paper is organized as follows. Section II presents the common security attacks that needs to be considered for secure hardware design. Section III illustrates the most common challenges for the existing security methods. Section IV presents a comprehensive review of hardware security methods. Section V explains categories of hardware attacks, followed by the conclusion in VI.

II. CLASSIC SECURITY ATTACKS

Electronic devices are the significant part of our daily life, and we will continue to depend on these devices more and more everyday. Exponential growth of the IoT devices and their ubiquitous connections are creating more security vulnerabilities. Now it is high time to be better prepared to tackle these attacks and design secure methods to protect and provide countermeasures for these attacks. To make this complex hardware security field a bit more organized, we have presented a summary of current hardware security approaches in Table 1, and categorization of the classic hardware attacks and countermeasures is shown in Fig. 2. In this section, basic security attacks types are described as follows:

A. HARDWARE TROJANS

Hardware Trojans mean inserting malicious modifications to the circuit. Internal complexity, scaling down of IC to the physical limits, and unspecified spaces available in the IC are some causes why Hardware Trojan (HT) detection is difficult. There are two types of threat models; the first one is manipulating the lithographic masks, which is when the foundry inserts a Trojan into the design in the form of addition, deletion, or modification of gates [55], [56].

The second type is when a malicious insider does not provide information about the HT to the validation team. However, HT can be inserted (modified) by a third-party IP design house. There are two types of Trojan detection techniques: one is called Invasive (and Semi-invasive), which requires costly, precision measurement equipment, and

components become unusable afterward. On the other hand, Non-invasive detection methods depend on external parametric and functional IC testing such as input and output pattern, delay, and dynamic leakage. Self-monitoring [57] and static verification [58] are two protections against rogue third party IP (3PIP) and insider attacks.

B. IP PIRACY AND IC OVERBUILDING

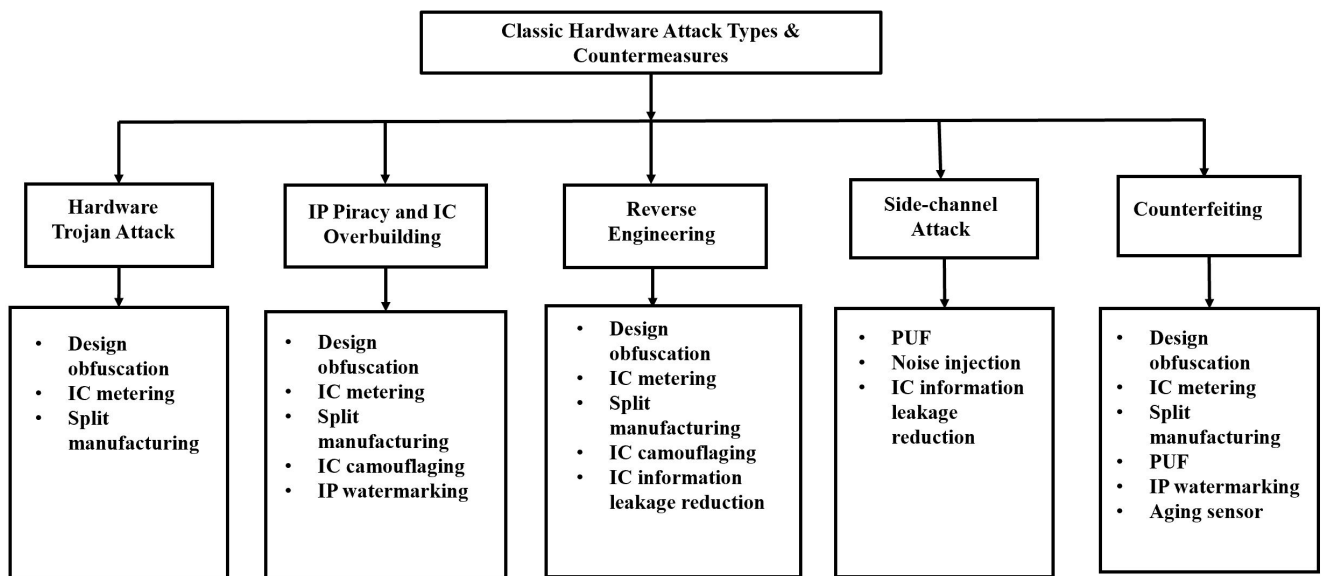
Modern System-on-Chip (SoC) and IC designs usually need different forms of IPs, for example, Register Transfer-Level (RTL) design (soft IP), gate-level netlist (firm IP), and physical layout (hard IP) [1]. The companies outsource IPs from trustworthy offshore design houses or foundries for SoC integration or IC fabrication to reduce design complexity, time-to-market pressure, and manufacturing cost. This globalization of outsourcing IPs can lead to various IP security threats, such as IP cloning, IP imitation, and inserting Trojans inside the IPs/ICs. This type of hardware attack happens if an outsider has excess to an IP or an IC and illegally claims ownership or steals the IP to overbuild and sell them [59], [60]. Attackers from the integration house or foundry can pirate 3PIP and use more than the amount of 3PIP instances allowed by the license, extracting information from the design layout, the IC design, and/or overbuilding it accordingly. Moreover, the fraudulent designs not only harm the reputation and earnings of the legitimate vendor but also seriously negatively affect important systems' performance and reliability because they contain harmful or backdoor logic that might leak private data or help override them [61].

C. SIDE-CHANNEL ANALYSIS (SCA)

Side-channels are instances of inadvertent information leakage where an attacker attempts to obtain data from a target computing system by exploiting technical flaws. Side-channel attacks attempt to extract secrets through a physical modality, for example, power consumption [62], electromagnetic (EM) emanations [63], photonic emissions [64], and

TABLE 1. Summary of attacks methods.

Attack name	No. of papers	References
Side-channel and covert channel	17	[3], [6], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]
Hardware Trojan	6	[35], [36], [37], [7], [38], [39]
IP piracy	4	[40], [37], [41], [42]
Counterfeiting	4	[43], [42], [44], [45]
Reverse Engineering	4	[43], [23], [42], [31]
Snooping	3	[46], [47], [48]
Denial-of-service (DoS) and Distributed denial-of-service (DDoS)	2	[49], [50]
Buffer overflow	1	[51]
Others	3	[52], [53], [54]

**FIGURE 2.** Classic hardware attacks and countermeasures [17].

acoustic noise of the system [65], etc. An attacker can use a different method to measure the switching power traces of an electronic device during its operation and after that attacker can use logical formulation to extract secret keys or secret data. Simple Power Analysis (SPA) and Differential Power Analysis (DPA) are methods for obtaining confidential information from a power SCA. Another technique for analyzing power channels is called Correlation Power Analysis (CPA) [66], which uses statistical models to calculate the correlation between the secret and the device's power usage when the secret is used for computing. Additionally, EM radiation during the execution of security-critical activities results in hardware vulnerability due to data leakage. EM radiation during the operation of security-critical processes also causes hardware vulnerability through leaking information [67], and many of these EM traces are helpful to extract secret keys. Leakage reduction, noise injection, key updates, side channel-resistance Physical Unclonable Function (PUF), and Secure Scan Chains are a few examples of protection against side-channel attacks.

D. COUNTERFEITING

Counterfeiting is frequently carried out by numerous companies involved in the semiconductor supply chain, such as new product suppliers or secondary (recycled) IC vendors. Selling used chip as new IC also fall into this category. An unauthorized imitation or forgery of the original semiconductor component is known as a counterfeit semiconductor component. One of the many organizations in the semiconductor supply chain, such as suppliers of new products or secondary (recycled) IC suppliers, frequently engages in counterfeiting. The ability to discern between fake and genuine ICs has recently improved because of technological advancements in 3-D packaging.

The manufacturers of authentic components lose money as a result of counterfeiting ICs in the market. The performance and dependability of the entire system are negatively impacted by the mediocre performance of counterfeit goods, usually in lower quality or older versions of the original product. The authentic provider's reputation is also harmed. Such counterfeit goods could potentially hamper the performance

of critical applications in healthcare, bank, cars, planes, and weapons control systems and cause lots of financial loss. It was calculated that US corporations suffered financial losses of between \$225 and \$600 billion in 2017 [68] due to design IP infringement. Some defense techniques for hardware attack are hardware metering and auditing (Tracking of manufactured integrated circuits after fabrication), IC Fingerprints or PUFs [69], [70] etc. Negative temperature bias instability (NBTI), hot carrier injection, and electromagnetic migration are some parameters that influence the lifetime of an IC, which can be estimated by the related sensor.

III. HARDWARE SECURITY CHALLENGES

Hardware security research methods face several challenges to be considered in the more advanced methods. In this section, the most common challenges are described as follows:

A. OUTSOURCING IP/INTEGRATED CIRCUIT

The fabrication process in the IC industry has seen several changes due to globalization in the semiconductor industry over the last decade [35]. The integrated circuit (IC) supply chain has become more globally diversified because of the time-to-market demand and the steadily increasing complexity of hardware designs [71]. Companies are also outsourcing IPs from third-party vendors to reduce the production cost in the IC supply chain. A diversified hardware supply chain creates diversified scope for vulnerabilities.

B. SECURITY FOR ENERGY LIMITED DEVICES/LIGHTWEIGHT DEVICES

Due to the growth of the Internet of Things (IoT) devices, reliable communication between them is becoming more of a concern. Although researchers have invested enormous efforts to solve this issue, most solutions place a heavy burden on energy, time, and complexity on IoT devices with limited energy [72]. Even though an analog-inspired encryption method for hardware security reduces energy consumption, latency, and complexity, but provides a moderate level of security. Therefore, a dynamic trade-off between security, power use, and performance needs to be established [22].

C. CONFIDENTIALITY OF DATA/DATA PRIVACY

One of the crucial aspects of hardware security is data privacy [36]. Due to the possibility that the trigger may not be included in the program code and data, vulnerabilities threatening the confidentiality of important signals, such as cryptographic keys, can go unnoticed in standard test scenarios. Outsourcing data storage which is common practice in the modern data-driven world also questions data security in the cloud.

D. SIDE-CHANNEL LEAKAGE REDUCTION

Another important hardware security challenge is to reduce side-channel leakage for example heat/temperature emission, EM wave emission, noise, vibration, etc. The physical

implementation of encryption, which is frequently employed to provide this confidentiality, is threatened by hardware side-channel attacks. Traditional hardware defenses against these side-channel attacks, including masking, typically include a significant energy overhead [23].

E. FLEXIBLE/RECONFIGURABLE/AUTOMATIC SECURITY

Flexibility and performance are well-known pairs among all design constraints in digital Integrated Circuits (ICs). Application-Specific ICs (ASICs) have little to no flexibility once the design phase is complete, but they provide the best performance for a specific set of activities. On the other side, reconfigurable computing with Field-Programmable Gate Arrays (FPGAs) enables the architecture to be modified at any anytime, but with a cost of slower speed, more power, and a larger area. The programmability of FPGAs further increases the attack surface from a hardware security standpoint [52]. As a result, finding a balance between flexibility and reconfigurability is also a challenging issue in the hardware security sector.

F. PACKAGING RELATED CHALLENGES

In order to gain high performance, shorter time to market, cost-efficiency, and small-sized ICs, advanced chip packaging is gaining attention, and as a result, packaging-related risks are also evolving now-a-days. For example, fan-in and fan-out wafer-level packaging (WLP) offers more opportunities to manufacture high-performance and small-sized ICs [73]. System on Chip (SoC) and System in Packaging (SiP) technologies are cost-efficient approaches for the industry to follow Moore's law [74]. Famous chip foundries such as TSMC and Samsung are also involved in advanced IC packaging. Therefore, fabless production of advanced packaging is inevitable and is a significant source of vulnerability to the packaging process. Moreover, various alterations can be easily implemented due to globalization and the complexity of the microelectronics supply chain. On the one hand, the globalization of the IC supply chain makes it more challenging for end-users or IP owners to guarantee that their components are guaranteed and tamper-free. On the other hand, the current detection methods cannot verify the internal materials of sample packaging due to the increased complexity of advanced IC packaging. In addition, adversarial changes in packaging parameters by a malicious manufacturer can result in undetected features, which can cause chip failure. As a result, chip packaging is now viewed as a severe hardware security issue [75].

G. SILICON-BACK SIDE VULNERABILITY

Data security has become a major issue because of the higher reliance on IoT edge devices that acquire confidential and private information and export it to the cloud network. Cryptography is used to provide network security for the IoT and edge devices by using two approaches: digital encryption algorithms for data protection and digital signature algorithms for device authentication. Several methods are used

to safeguard this SoC module, for example, the front side of the SoC has metal layers and shielding meshes that block incident photons' paths and shield the ICs from optical attack. However, the silicon substrate of the gadget, which is its backside, lacks any such safeguards [76]. An attacker can effortlessly track and extract security-sensitive data and information by laser/optical probing, fault injection, and side-channel attacks. But in practice, flip-chip packaging is frequently employed in these IoT devices because they provide a smaller form factor even with more functionality and lower power. As flip-chip exposes the Si substrate to the surface of an IC chip, leading to a considerable security gap. Flip-chip can be achieved with packaging that is narrow and practically chip-scale which make them ideal for IoT devices. Therefore, an attacker can effortlessly track and extract security-sensitive data and information by laser/optical probing, fault injection, and side-channel attacks. As a result, a Si-backside of an IC chip, especially in Flip-chip packaging leads to a serious security hole [32].

IV. HARDWARE SECURITY APPROACHES

Several hardware security approaches are presented in the existing literature to consider several domains. A comparison table between the existing approaches, is shown in Table 2. Guo et al. [3] propose a language-based framework, Quantified Information Flow- Verilog (OIF-Verilog), to evaluate the trustworthiness of a hardware system at a register transfer level (RTL). QIF model applies information-theoretic metric to measure the flow of sensitive information in hardware design. The names of the QIF metrics are Remaining Uncertainty (RU) and Accumulated Remaining Uncertainty (AR). This method solved scalability and hardware overhead issues, and able to detect logics that are vulnerable and malicious.

Gogniat et al. [49] focus on the challenges of developing secure embedded systems using reconfigurable hardware and introduce a security architecture for embedded systems (SAFES). Three main concepts underpin the SAFES architecture: 1) reconfigurable security primitives; 2) reconfigurable hardware monitors; and 3) a hierarchy of security controllers at the primitive, system, and executive levels. The targets of the proposed architecture are to continuously monitor the system's operation to detect abnormal behavior and to use reconfigurable hardware to provide various levels of protection and performance. The flexibility of the given solution makes it possible to address the security issue while also defining an energy-efficient solution, as demonstrated by results from the AES and RC6 algorithms. The main drawback of this method is that some parameters need an investigation, such as the overhead for security mechanisms in reconfigurable hardware and how to make security commonplace in embedded systems.

Nagata et al. [6] propose IC chip-level hardware security using the backside buried metal (BBM) wirings to achieve avoidance, detection, and resiliency against electromagnetic and laser attacks. Secure IC packaging is essential so that no third party can compromise IC security throughout the

supply chain. This article focused on IC-chip-level protection against physical attacks for cryptographic circuits in IoT applications, which is one of the common hardware security issues. The physical structures for secure IC chips are also explored to protect them against multimodal side-channel attacks. The Backside Buried Metal (BBM) wirings in a Si substrate are incorporated with the frontside Complementary Metal Oxide Semiconductor (CMOS) circuits to prevent IC chip-level attacks.

Naveenkumar et al. [40] present anti-SAT-based structural obfuscation for the C17 benchmark with iterations. Compare the number of iterations necessary to find the correct key between the proposed design and conventional Anti-SAT with the locking design. SAT attacks threaten the logic locking security by decrypting the appropriate key within a short period of time, even with many keys. So, this study suggested XOR/XNOR logic locking anti-sat with structural obfuscation concept, enhancing hardware security with an iteration perspective. The number of iterations required to extract a proper key by SAT attack is proportional to the key size in the circuit having an Anti-Sat block.

Eslami et al. [35] propose a security metric and an assertion selection flow that leverages Cadence JasperGold Security Path Verification (SPV). This study depicts how to use one of the existing verification assets, named assertion, which can be applied as an online monitor security checker for hardware Trojan. This study also proposed a security metric called security coverage to evaluate the efficiency of online checkers in detecting Trojans, considering the amount of overhead imposed on the circuit. Finally, it proposed an assertion selection flow to leverage Cadence JasperGold Security Path Verification (SPV). This method can detect the Trojan within its reachable area which is on average 6.8% of the total node in the circuit, no matter how hard the Trojan is.

Reimann et al. [36] present a study to reformulate an approximation technique and propose a tool named QFlow to identify data leakages in hardware designs, thus, facilitating a security-aware design process. Most information flow techniques depend on the non-interface approach assuming untrusted targets should not be affected by the source's data. An approximation method was necessary to avoid complex quantification techniques to solve this issue. This study proposed a tool named QFlow to identify data leakages in hardware designs and used multiple open-source benchmarks to show the efficiency and flexibility of QFlow, thus facilitating a security-aware design process.

Rostami et al. [17] discuss the hardware-based attack, state-of-the-art defenses, and the metrics used to evaluate the defenses. As the IC is scaling down, its cost and complexity are also increasing. As a result, IC production is shifted overseas, making the IC production supply chain more prone to hardware security attacks. Moreover, critical control and communication functions assume that the hardware platform they are implemented on is secure, but this assumption is not always valid. For example, reference [4] reported that backdoors could be used to reostat weapon control systems,

transportation systems, and nuclear power plants. Counterfeit electronics, a security vulnerability in hotel keycards, and a security vulnerability in parking payment cards are some examples of practical hardware attacks.

Ellicott et al. [72] propose and demonstrated an innovative cryptographic approach for low-power, complex, and low-latency IoT wireless communications. Energy and latency reduction for the system was done by analog encryption that takes place inside the Radio Frequency (RF) front-end of the radio. It removes the need for bit-by-bit digital encryption for much of the transmitted data. The goal of the system is to keep an eavesdropper's Bit Error Rate (BER) close to 0.5, as this left them in the dark regarding the transmitted data.

Yu et al. [37] propose HW2VEC, an open-source (python library) graph learning tool for hardware security to provide an automated pipeline for extracting graphic representations from a hardware design in either Register transfer level (RTL), and Gate-Level Netlist (GLN) levels. HW2VEC users can automatically convert non-Euclidean hardware designs to Euclidean graph embeddings to solve problems. HW2VEC also enables the transformation of hardware designs into different graph representations, including the Data Flow Graph (DFG) and Abstract Syntax Tree (AST). This method can be used to assess the similarities between circuits and thus can be a countermeasure for IP piracy at both Register Transfer Level (RTL) and Gate-Level Netlist (GLN) levels. It also automatically transforms non-Euclidean hardware designs into Euclidean graph embedding for solving their problems. By adjusting the Trainer to train the GNN models for node classification, HW2VEC can be adapted to localize the HTs or hardware bugs that exist in the hardware designs. Meanwhile, this method has some drawbacks such as HW2VEC spending in converting the raw hardware code into ASTs is, on average, 1.98 seconds. HW2VEC also performs worse in DFG extraction but better than the 6-9 months require for manual feature engineering.

Arias et al. [38] propose a framework named RTSC that can perform security analysis on designs and integrate security features directly into the HDL code (watermarking and logic locking), a feature that commercial EDA tools do not provide. If a designer wants to watermark a finite state machine, it needs lots of complex steps or if a designer wants to obfuscate a logic within the HDL design also needs lots of complex steps. This research also showed how to use RTSEC to automatically include security mechanisms in RTL designs: watermarking and logic locking. RTSEC performs a series of inference checks to extract features of interest in the design after first decomposing the given HDL into an abstract syntax tree (AST). User-specified security upgrades can then be incorporated into the design via RTSEC. After that, it generates new HDL code for the design that incorporates the improvements without altering its semantics. But restricted to see if the tools can infer finite state machines.

Wei et al. [7] introduce a chopper-stabilized amplification technique as part of a fully differential temperature sensor to detect HT, based on on-chip power monitoring. The signal

quality is limited by flicker noise when temperature sensors are used to measure the power dissipation of surrounding circuits to detect malicious circuits in the design, because the slowly shifting temperature gradients imply that the information of interest is at low frequencies. To improve signal quality, this study suggested a two-stage trans-impedance amplifier layout with a chopping method for flicker noise reduction by modifying the differential input currents from the deployed bipolar junction transistors used as temperature sensors. In this method, a signal-to-noise ratio has been significantly improved, but 3rd harmonic (HD3) degradation occurs.

Dobrykh et al. [20] propose a shielding mechanism using opaque metal to prevent a far-field attack on Radio Frequency Identification (RFID) chip. RFID is widely used as a contactless data exchange for the billing system. One theft in RFID-based communication systems is when approaching a victim with a card reader. The physical principle of operation is to place a high-index dielectric structure next to the sub-wavelength aperture, making it electromagnetically larger, thus supporting electric field leakage. Introduced smart cover could practically eliminate the capability of the readout without authorization.

Dutta et al. [43] propose a run-time reconfigurable camouflage logic technology based on Ferroelectric Field-Effect Transistor (FeFET) and demonstrate it in silicon gate-level camouflaging by implementing three Boolean logic functions NOR, NAND, and XNOR by using threshold voltage (VT) programming of FeFET. The proposed method obfuscates design IPs from untrusted foundries by protecting against reverse engineering and counterfeit threat. This method can obfuscate design IP from the zero-trust foundry and can achieve overhead reduction such as area, power, and delay. However, the Threshold Voltage Defined (TVD)-based circuit topology used in this article does not directly address the state-of-the-art satisfiability problem (SAT) attacks.

Bernard et al. [52] introduce the dual-function asynchronous polymorphic logic gates which have functionality selectable by a change in supply voltage. This work proposed dual-function asynchronous polymorphic logic gates whose functionality can be selected by changing supply voltage and verified in silicon using the TSMC 65nm process. Asynchronous Polymorphic Gates were designed to achieve the "hidden" reconfigurability without the specific area and security overhead of Field Programmable Gate Array (FPGA) hardware implementations. This method requires less timing analysis and achieved a 40% reduction in transistor count for most asynchronous polymorphic gates designed using this method. But this method has a power tradeoff.

Kumar et al. [21] propose a method that used JasperGold an industrial strength model checker to investigate the design of different units of 1200 processors (5-stage pipeline design) for security threats or to detect hardware design error. In this work, security-critical features were created through iterative refinement based on an understanding of processor specifications obtained from manuals or earlier research. It intends to write properties in System Verilog Assertions (SVA) format

so that a model checker tool could easily use them to check their properties. The suggested method thoroughly examines the model checker-provided counterexamples of the properties' failures to find security flaws. The proposed method explores a wider section of the design state space compared to the simulation-based method. However, this work did not consider the coverage issue for all design-related security matters.

Dessouky et al. [53] propose a consolidated hardware-assisted security extension (CHASE) that supports various levels of security at runtime and can be configured at runtime. CHASE can be set appropriately at runtime depending on the required security level and the system's real-time availability or functionality needs, enabling the calibration of the security vs. performance trade-off. By categorizing different classes of attack that may target embedded systems, this work evaluates which attacks can be detected on-device and which require more sophisticated policy-checking at the third party. In this work, a centralized and reconfigurable defensive mechanism was presented that can be changed at runtime to offer various security services (and various security assurances against various attack classes) at the expense of various performance overheads. CHASE offers greater attack coverage than current defenses and is applicable to timing-critical systems. But the hash engine incurs 50% of the overhead in this method. Invasive/semi-invasive physical attacks are out of scope in this work.

Patnaik et al. [41] design a Giant spin hall effect (GSHE)-based primitive for IP protection utilizing Logic locking and camouflaging. The main idea of this research is to ensure hardware security in terms of imprecise computing. Compared to conventional gates, this technique can prevent Reverse engineering (RE) attacks on the chip's design IP without requiring the semiconductor designer to change the underlying netlist. Probabilistic SAT (PSAT) can successfully clear up polymorphic obfuscation on more minor scales. However, the success rate is constrained at larger scales. Proposed method is effective for SAT attacks and has inherent resilience of the GSHE device against side-channel attacks.

Hardware security is one of the major research topics that significantly relies on CMOS-based ICs, and the defense and attack mechanisms are mainly based on CMOS-based structures. Divyanshu et al. [23] investigate several new magnetic tunnel junction (MTJ)-based structures for potential logic locking applications in hardware security systems. This work also performs logic locking application on different MTJ-based devices to test the viability and applicability of such devices even in the presence of process variations for future security applications compared to CMOS device. MTJ-based structures provide an extra layer of protection, but it has area overhead.

Ho et al. [24] propose a Multicore power balancing technique by configuring data routing paths in the 9-core Network-on-Chip (NoC) based platform as a countermeasure for differential power analysis (DPA) attacks. The proposed method is flexible to adjust the cores' positions, the routers'

locations, and the clock frequencies on each node to balance power dissipation and prevent DPA. The Normalized energy deviation (NED) [20] considerably decreases by 52% when there are 5 functioning routers, or 1 core and 5 routers, as opposed to the simple (1 core + 1 router) AES operation. Similarly, the NED considerably decreases by 73% when the running core count is increased from 1 to 4, or (4 cores + 4 routers). The proposed method is reconfigurable, hence is able to support different algorithms other than AES and can perform multitasking. It is also energy efficient. However, the NED for the Advanced Encryption Standard (AES) operation is unacceptably high at 0.21 for power-on (1 core + 1 router).

Thnagam et al. [42] propose a logic locking technique to protect the IC from the attackers by obfuscating the attackers, because a key is generated by using the Pseudo Random Number Generator. The logic locking technique was applied to the Montgomery algorithm to test the proposed logic locking method. The purpose of this research is to stop IC from being pirated using techniques like PUF-Binding, hardware metering, camouflaging, and obfuscating. The results of the experiment demonstrate that the proposed technique considerably decreases the area and delay. This method has less area and delay overhead, but it has high switching activity and excessive power consumption.

Gong et al. [25] propose a secure scan chain technique using retention loss characteristics of memristors devices for hardware security. The followings are the contributions of this article: firstly, this work proposes a new design to achieve scan obfuscation on analog-based memristor devices. Furthermore, it develops an analytical scheme to recover scan responses. Secondly, this work enhances obfuscation performance by creating statistical models to serve as a guide for figuring out the vital design elements, including the memristor refresh frequency. Finally, circuit implementations are shown, demonstrating the memristor-based obfuscation method with minimal hardware overhead and timing and power restrictions. The existing scan chain structures can easily adopt the proposed approach without redesigns or modifications. By comparing the test key, it distinguishes between authorized and unauthorized users. For unauthorized users, it introduces true randomness from degraded memory content to conceal scan responses. By comparing the test key, it distinguishes between authorized and unauthorized users. For unauthorized users, it introduces true randomness from degraded memory content to conceal scan responses. This method is suitable for resource-constrained implementations. However, a large array could increase the overhead and slow down the memory degradation, and it has very little hardware, timing, and power overhead.

Chittamuru et al. [46] introduce a novel framework called SOTERIA that employs architecture-level improvements and process variation-based authentication signatures to safeguard data in Photonic Network-on-Chip (PNoC) architectures from snooping attacks. Without making major changes to the architecture, the suggested framework can be easily implemented in any Dense-Wavelength-Division-

Multiplexing (DWDM)-based PNoC that already exists. To protect data from eavesdropping attacks in photonic waveguides, a circuit-level PV-based security enhancement scheme was proposed. An architecture-level reservation-assisted security enhancement scheme was also suggested to enhance security in DWDM-based PNoCs. The proposed method combines the circuit- and architecture-level schemes into a comprehensive framework called SOTERIA and analyzes it on the Firefly [8] and Flexishare [9] crossbars. SOTERIA has just modest overheads of up to 10.6% in average latency and up to 13.3% in the Energy-Delay Product (EDP), significantly improving the hardware security in dense-wavelength division-multiplexing (DWDM) based PNoCs.

In order to bridge the frequency difference between reconfigurable hardware and the core processor, Mao et al. [26] suggest an architecture named REHAD, which also made use of the hardware parallelism concept. REHAD is an architecture that uses a low-frequency reconfigurable hardware attack detection module for cache side-channel attacks detection. Microarchitectural timing attacks known as “Cache Side-Channel Attacks” (CSCAs) primarily target shared caches. Hardware detection modules can be reconfigured by means of a trusted software kernel, to adapt to the level of threats and to detect new attacks. The proposed architecture can fill put the frequency gap with reconfigurable hardware and core processor. A simple heuristics, implemented in the trusted software kernel can efficiently detect CSCAs. After that REHAD has been integrated into the ORCA softcore RISC-V on FPGA. REHAD can detect two common side-channel attacks namely Flush+Reload and Prime+Probe successfully. REHAD architecture can detect only 5 out of 8 lines of the victim’s monitored memory can be detected by the malicious process. Performance and area overhead for the full REHAD architecture are not analyzed in this work.

Kusto et al. [44] propose a femtosecond laser-induced printing using all-dielectric silicon nanoparticles (NPs) on polypropylene substrate as a novel fabrication method of anti-counterfeiting labels. The proposed method offers the unique distribution of nanoparticles (NPs) and resonant optical properties of silicon nanoparticles, such as Raman scattering and far-field scattering, as forensic security features for security tags. Moreover, this work also demonstrates that the laser-induced printing technique allows the creation of arrays of NPs arranged in arbitrary texts, symbols, and images. It is suitable for flexible PUF tags fabrication, inherently fast and easy-to-implement method. Flexible protective tag creation using All-dielectric silicon Nano Particles (NPs) on polypropylene substrate is a promising option to prevent counterfeiting products.

Ma et al. [47] propose a practical scheme named Dam to mitigate data-oriented attacks with tagged memory based on hardware. The proposed approach uses the idea of tagged memory to break data-flow stitching and gadgets dispatcher of generating data-oriented attacks. Dam was implemented by extending lowRISC, which is a RISC-V based SoC

(System of a Chip) that implements tagged memory. This method eliminates data-oriented attacks by paying attention to memory access. Two essential rules for security enforcement were used in the proposed method. Rule 1 indicates that the Data pointers must point to valid memory addresses, which is named trusted set of data pointer objects (legitimate DPO). Rule 2 states that crucial variables must be described by legal memory access procedures, which they referred to as valid decision variables (legitimate DV). This method provides source compatibility, strong security, and low cost.

Montoya et al. [27] propose a novel countermeasure called dynamic encoding using a shift register to balance power consumption without precharge at the algorithm level and require at most 8 random bits at each encryption. Protecting lightweight algorithms in resource-constrained devices is challenging because they require up to thousands of bits of fresh randomness at each encryption and create power and area overhead. To compensate mismatch in the propagation times and capacitance due to silicon implementation, a random dynamic encoding can change the encoding table at each execution. This method is expensive and has a smaller area and power overhead. Moreover, this suggested method is suited to secure lightweight shift register-based security primitives.

Mohammed et al. [28] propose a non-invasive approach that investigates Hardware Intrinsic Attack Detection in IoT (HIADIoT) devices through the power profile of IoT devices in different modes of operation utilizing machine learning algorithms. Random forest algorithm used in this research, which can correctly classify 95.5% of the data point and recognizes potential Hardware intrinsic (HI) attacks. The suggested method can correctly classify 95.5% of the data point and recognizes potential HI attacks. But this method is not applicable to the HI in IC.

Ding et al. [29] propose a method that achieves entropy for ring oscillator-based True random number generator (TRNG) and Physical unclonable function (PUF) by using dynamic Threshold switching (TS) variations and static leakage current mismatch. By digitizing the stochastic oscillator periods induced by thermal noise during the TS process, the proposed TRNG presents excellent immunity against power noise attack (frequency range up to 400MHz, and amplitude range up to 600mV), according to the NIST SP800-90B and NIST SP800-22 tests confirmation. Moreover, an autocorrelation test was done to check the randomness of both TRNG and PUF.

Lin et al. [30] propose a Multiphysics simulation framework that efficiently generates time-domain EM traces at any virtual probe above the surface of the silicon substrate by modeling cell-level power, chip logic functionality, and layout geometry. In order to predict the most vulnerable leakage location, ML-based automatic POI (point-of-interest) identification algorithm was also proposed in this work. The proposed method can identify near-field electromagnetic (EM) side-channel leakage 0-100x faster than a conventional correlation-based side-channel simulation approach.

This work also suggests that the highest region of EM leakage is caused by the ringside metal and I/O portion (i.e. Vdd/Vss pads) of the chip.

In order to get a smaller form factor and high efficiency, 3D integration technology has got high attention from researchers in the past few years. But 3D integration technology also brought some new challenges to the hardware security field. Zhu et al. [31] present a wet-etching-based 3D package structure to prevent EM radiation from the backside of the die. The proposed method uses two dies which are embedded in the substrate trench, and one die flips down and connects with the two dies via the redistribution layer. The embedded die is surrounded by metal wire, which constructs a metal shielding network and can effectively prevent electromagnetic radiation leakage from the die. The metal shielding network can also act as an invasive attack detection circuit. Using wet-etching-based 3D package structure, effectively reduce EM radiation. The author did not show any comparison for overhead issues.

Flip-chip packaging creates a significant security gap that opens the door to new backside physical security attacks. CMOS detector circuit combined with BBM copper (Cu) meander to protect the cryptographic circuits against physical security attacks from vulnerable backside Si substrates is proposed Miki et al. [32]. To protect the cryptographic circuit from physical security attacks from vulnerable backside Si substrate, the BBM Cu meander was monolithically combined with CMOS detector circuits. This BBM circuit approach prevents the backside SCA from revealing the secret information of cryptography because the BBM filters out the internal substrate noise and prevents probing from acquiring it.

Lorandel et al. [48] propose a hardware architecture that is introduced and demonstrated, showing the efficiency of traffic modeling and the ability to detect two different forms of hardware attacks on the I2C, the most popular serial communication bus. The idea is to detect potential attacks relies on the characterization of the wired communication traffic occurring at the interface of the IoT devices.

Manssour et al. [39] propose hardware support for instruction replay, consisting of a small extension of the instruction set (one new instruction), a few detection elements (mainly registers and comparators), and minor modifications of the processor control. A small 32-bit RISC processor is used to add hardware support for instruction replay. Hardware support for instruction replay allows the automatic and parametrized replay of sequences of instructions. Protection efficiency is similar to the software replay solutions, but with shorter execution time and code size required. This process allows the automatic and parametrized replay of sequences of instructions. Only the data flow is protected, not the control flow. It has a faster execution time. 75% of faults can be detected by using this method and has small area overhead.

Logic locking is a prominent key-driven technique to protect the integrity of integrated circuits in the IC supply chain.

But, recently introduced machine learning-based attacks are able to recover secret keys without having access to an activated circuit. A study by Sisejkovic et al. [54] propose a deceptive multiplexer (D-MUX) based logic locking scheme which is resilient against structure-exploiting machine learning attacks. This study also analyzed a machine learning-based logic locking scheme. The main idea is that each path through a MUX must have the same probability of being true or false for a D-MUX to function correctly. The limitations of logic locking in the context of learning-based attacks were also addressed in this work. This method is able to defend against all recognized learning-based attacks.

Bhade and Sinha [33] propose an architecture-agnostic approach using hardware performance counters at run time and thread level to protect against Cache Side Channel Attacks. The proposed method reduces the false positives by 48% compared with system-level approaches. The proposed method uses a micro-architecture profiler and hardware event counters. Either manually or automatically sensitive thread of the user application needs to be marked first. Micro-architectural event counts will be observed and compared with the predetermined threshold value, and the attack will be detected if the count value exceeds the threshold value. The suggested method works better in embedded systems when processor cycle time is a limited resource. The proposed method can provide greater precision and dependability than system-level monitoring. It Works better for low-frequency applications.

A cache side-channel attack is a CPU-shared resource-related vulnerability. Most previous detection mechanisms were designed to monitor cache events in the CPU carefully. However, this trend has changed; for example, the PRIME+ABORT attack takes advantage of the Intel TSX hardware instead of accessing the cache to measure access time. So, it is necessary to find out other hardware-related cache side-channel detection methods. Kim et al. [77] present a technique called the PRIME+ABORT detector, which records selected hardware event counts using the hardware counter during AES decryption and the PRIME+ABORT attack. In this paper, count for the selected event was performed every second. In this work, AES decryption was run 11200 times, and PRIME and ABORT attack was run 8500 times. The proposed method can achieve a 99.5% success rate with only 0.3% performance overhead during 30 minutes of execution time. Cache side-channel attack detector requires careful customization on which it will run to guarantee an acceptable detection performance.

A Plethora of applications in modern IoT devices requires synergically participation among different individuals and organizations for trusted communication. As a result, attack on IoT devices is increasing alarmingly. There are two typical attack phases: firstly, attackers exploit vulnerabilities in as many devices as possible and then turn them into bots. Secondly, the attacker dispatches as many requests as possible to the victim device and exhaust the target device's computing resources called Distributed Denial of Service (DDoS).

A mitigation technique is proposed Brasilino et al. [50] to improve resilience against DDoS attacks in IoT devices. The proposed architecture uses a CoAP accelerator that cooperates with the device's CPU to alleviate the IoT SoC's CPU burden of handling incoming and outgoing CoAP messages. When the CoAP is activated, it will handle messages without CPU interaction. As a result, the CPU will be free to execute helpful work. A prototype of the proposed architecture was also implemented in SoC FPGA, and an evaluation was also carried out to see if the objective was satisfied. Advantages: a Distributed Denial of Service (DDoS).

Counterfeit IC is a great threat to the modern IC supply chain. There are a few counterfeit IC identification techniques proposed already, such as implanting Radio Frequency (RF) antenna into the IC structure or using Scanning Microwave Impedance Microscopy (sMIM) to measure high spatial resolution material and device characterization. THz range stays between the microwave and infrared range. THz bridges the electronic and optical portions of the spectrum, but it suffers from poor spatial resolution. A work by True et al. [45] presents an advanced metrology method to detect authentic vs counterfeit IC. The proposed method uses THz point spread function to improve spatial resolution. This high-fidelity model compares the digital structural and spectral properties in unsupervised deep learning (DL). This is an automated counterfeit detection method and it requires minimal to no manual labor. The method has a drawback, the overall spatial and spectral resolution of THz needs to be improved.

Buffer overflow attack is one of the frequently listed hardware attack in the vulnerability database [78], [79]. When a microcontroller gets filled with more than its assigned capacity, the overflowing data replaces the return address location, which alters the original address to which the function would have been required to return. Using Verilog codes, a simple and efficient mitigation technique is proposed by Sayeeshwari and Prabhu [51]. The proposed method was also simulated in Vivado. A hardware-integrated system for determining the memory capacity's bounds was developed to provide a simple countermeasure to buffer overflow attacks. The proposed method allows both passcode verification and checking the buffer memory's boundary limits. Allows usage of a minimalistic number of logic gates as an encryption method.

V. HARDWARE ATTACK RELATED APPROACHES

The recent hardware attack related approaches are categorized in Fig. 3 and summarized weakness and strength, is shown in Table 3.

A. CLASSIC HARDWARE ATTACK

1) PACKET HEADER ATTACK BY HARDWARE TROJAN

Kulkarni et al. [82] model an HT that can alter the destination address field of specific Network-on-Chip packets and is mounted on the input buffers of NoC routers. Then it was simulated in a Tiled Chip Multicore Processors (TCMP)

whose effect leads to packet drop upon reaching a wrong destination tile. This paper also finds a suitable area within the NoC router where an HT can install itself and change packet content without interfering with its fundamental functionality. Proposed HT can severely affect the miss penalty of L1 caches and even stop an application by blocking an instruction. But, only applies to situations where the cores perform low injection rate programs.

2) TAMPERING ATTACK ON AES CORES WITH HARDWARE TROJANS

The study by Jain and Guin [83] describe a unique tampering technique using a sequential hardware Trojan to modify and obtain the secret key from the AES core. The proposed method depends on When an untrusted foundry inserts a hardware trojan in the netlist. Proposed Trojan activation can modify the input data of a particular round by preventing key-dependent computation of the previous round. Then the proposed sequential hardware Trojan helps the attacker to collect this modified data at the input of an internal round. In the proposed method, Trojan allows an attacker calculate the last round subkey using the observed ciphertext. An algorithm to recover the original key from the subkey from the previous round was provided in this paper.

Since the attacker only knows the trigger condition generated from the input plaintext and applies it repeatedly, it is very challenging to identify such Trojans using logic testing. In this method, overhead is caused by the payload because it needs 128 OR gates.

3) HARDWARE TROJANS ATTACK ON ANALOG/MIXEDSIGNAL ICs

Designing HTs for analog ICs is highly difficult because it is hard to meet all the requirements for a successful HT. Firstly, as analog signal pathways are usually susceptible, it is challenging to design stealthy HT because doing so could result in non-negligible performance deterioration. Secondly, because analog designs often have few components or can be easily separated into sub-blocks or stages with a few components apiece; it is challenging to create tiny footprint HTs that will escape optical reverse engineering. Finally, as the researchers can extract multiple information-rich measurements from any analog IC, testing will likely be able to distinguish HT-infected and HT-free instances. Despite all difficulties mentioned above, designing an analog Trojan is possible.

The paper by Elshamy et al. [84] propose a possible HT attack scenario in which the Trojan does not reside inside the analog circuit of the victim. However, it is located on the independent digital circuit on the same die where it was triggered. Its payload is only applied to the analog circuit after being transmitted via the common test infrastructure and the analog circuit's test interface. This HT attack uses the dense digital circuit to conceal its footprint successfully, making it impossible to detect in the analog domain.

TABLE 2. A comparison between the existing hardware security methods.

Ref.	Description	Attacks	Advantages	Drawbacks
[49]	This study focuses on issues related to building secure embedded systems using re-configurable hardware and proposes a security architecture for embedded systems (SAFEST) based on three main ideas: 1) reconfigurable security primitives; 2) reconfigurable hardware monitors; and 3) a hierarchy of security controllers at the primitive, system, and executive level.	• Hijacking • Denial-of-service (e.g., draining of battery or causing a battery to overheat) • Extraction of secret information (e.g., user's phone book)	• Adjusting the security measures dynamically for changing conditions (i.e., attacks, performance, power). • SAFEST can continuously monitor the system to detect anomalous behavior.	• High complexity
[3]	This study presents a new hardware description language called QIF-Verilog, an extension of Verilog, that uses taint propagation to trace the information flow of a specified type of signal at the register transfer level.	• Information leakage vulnerability	• It Solves scalability and hardware overhead issues, able to detect vulnerable and malicious logic. • Developers without a background in security can quickly assess the design's reliability by intuitively labeling the sensitive signal using the QIF-Verilog.	• This method works well if the system leaks information through logical output only.
[6]	This article focuses on IC-chip-level protection against physical attacks for cryptographic circuits in IoT applications, which is one of the many hardware security issues. The physical structures for secure IC chips are also explored to protect them against multimodal side-channel attacks.	• EM emission • Laser injection	• The Backside Buried Metal (BBM) wirings in a Si substrate were incorporated with the frontside Complementary Metal Oxide Semiconductor (CMOS) circuits to prevent IC chip-level attacks. • On-chip monitoring circuit was designed for sensing attempts and detecting attacks.	N/A
[40]	It proposes Anti-SAT-based structural obfuscation for the C17 benchmark with iterations. Compare the number of iterations necessary to find the correct key between the proposed design and conventional Anti-SAT with the locking design.	• Piracy • Counterfeiting • SAT attacks	• The proposed algorithm can resist around 32768 iterations against SAT attacks as compared to conventional ones.	• High power consumption • Increased area overhead • Increased delay overhead
[35]	This study proposes a security metric and an assertion selection flow that leverages Cadence JasperGold Security path verification.	• Hardware Trojan	• It is pragmatic since it does not rely on the hardware trojan activation mechanism. • The simplicity of using it without complex procedures. • Scalability.	• Normalized numbers above one might not fall within the circuit's noise margin and have an impact on performance. • On average, 6.8% of the circuit's total nodes are covered by the proposed method.
[36]	This study reformulates an approximation technique and proposes a tool named QFlow to identify data leakages in hardware designs, thus, facilitating a security-aware design process.	• Hardware Trojan	• Higher detection rate compared to the previous methods. • Less complex • Non-experienced users can also identify data leakages in hardware design.	• Computational time did not improve significantly.
[37]	This research proposes HW2VEC, an open-source (python library) graph learning tool for automating hardware security for hardware security practitioners to extract graph representations from a hardware design in either (RTL) and Gate-Level Netlist (GLN) levels.	• Hardware Trojan • Intellectual Property Piracy	• It can be used to assess the similarities between circuits and thus can be a countermeasure for IP piracy at both RTL and GLN levels. • It can automatically transform non-Euclidean hardware designs into Euclidean graph embeddings for solving their problems. • HW2VEC can be adapted to localize the HT(s) or hardware bug(s) that exist in the hardware designs by adjusting the trainer to train the GNN models for node classification.	• High time overhead • HW2VEC spends on average 1.98 seconds in converting the raw hardware code into ASTs. Although, [80] spends 1.37 seconds on average per hardware code. • HW2VEC performs worse in DFG extraction, but better than the 6-9 months requirement for manual feature engineering.

TABLE 2. (Continued.) A comparison between the existing hardware security methods.

[38]	It proposes a framework named RTSC that can perform security analysis on designs and integrate security features directly into the HDL code (watermarking and logic locking), a feature that commercial EDA tools do not provide.	Identifying finite state machines from the design	It can automatically include security mechanisms in RTL designs: watermarking and logic locking.	It is restricted to knowing if the tools can infer finite state machines.
[7]	A chopper-stabilized amplification technique was introduced as part of a fully differential temperature sensor to detect HT based on on-chip power monitoring.	Malicious circuits	The signal-to-noise ratio has been significantly improved	It has 3rd harmonic (HD3) degradation.
[20]	It proposes RFID chip shielding using opaque metal to prevent Far-field attacks.	Far-field attacks	Smart cover could practically eliminate the capability of the readout without authorization.	- Only a minor dependence on an end-user. - Aperture size between 2cm² to 12cm² allows a reading range of the authorized tag up to 2m.
[43]	It proposes and implemented a run-time re-configurable camouflage logic technology based on a FeFET.	- Reverse engineering - Counterfeiting threat	- It obfuscates design IP from a zero-trust foundry. - It reduces overhead in area, power, and delay.	The TVD-based circuit topology used in this article does not directly address the SAT attacks.
[52]	The dual-function asynchronous polymorphic logic gates in this research have functionality selectable by a change in supply voltage.	It can thwart several longstanding hardware attacks in the state of the art.	- Hiding functionality 40% reduction in transistor count for most asynchronous polymorphic gates designed using this method. - It requires less timing analysis.	Power-area tradeoff
[21]	JasperGold an industrial strength model checker was used to investigate the design of different units of or 1200 processors (5-stage pipeline design) for security threats.	Hardware design errors	It can explore a wider section of the design state space compared to the simulation-based method.	This study did not consider the coverage issue for all design-related security matters.
[53]	This research proposes a consolidated hardware-assisted security extension (CHASE) that supports various levels of security at runtime and can be configured at runtime. CHASE can be set appropriately at runtime depending on the required security level and the system's real-time availability or functionality needs, enabling the calibration of the security vs. performance trade-off.	- Code injection attacks - Control-flow hijacking attacks - Remote software attacks	- Covering a larger attack space than existing defenses. - It is applicable to timing-critical systems.	- Hash Engine incurs 50% of this overhead. - Invasive/semi-invasive physical attacks are out of scope in this work.
[41]	It proposes a design of a giant spin-hall effect (GSHE)-based primitive for IP protection by means of layout obfuscation.	IP protection (by means of logic locking and camouflaging).	- It can thwart most existing SAT attacks. - Inherent resilience of the GSHE device against side-channel attacks.	Regarding delays, the GSHE version is mediocre compared to standard (i.e., unprotected) CMOS implementations similar to previous literature.
[22]	It proposes and implements an adaptive second-order threshold implementation masking of Trivium to tune the security level dynamically during the execution of Trivium without any impact on its performance.	Side-channel attacks	Reduced power consumption	Small area overhead
[23]	It performs logic locking applications on different MTJ-based devices to test the viability and applicability of such devices even in the presence of process variations for future security applications compared to CMOS devices.	Reverse engineering power attacks	- It adds an extra layer of security. - It is more challenging to reverse engineer multi-staged logic locking units because the complicated switching mechanism used needs to be decoded.	High area overhead
[24]	A multicore power balancing technique was proposed by configuring data routing paths in the 9-core NoC-based platform as a countermeasure for Differential Power Analysis (DPA) countermeasures.	DPA	- Reconfigurable, hence it is able to support different algorithms other than AES. - Performing multi-tasking - Energy efficient	The Normalized Energy Deviation (NED) for the AES operation is unacceptably high at 0.21 for power-on (1 core + 1 router).

TABLE 2. (Continued.) A comparison between the existing hardware security methods.

[42]	A logic locking technique is developed to protect the IC from the attackers by obfuscating the attackers, because the key is generated by using the Pseudo Random Number Generator.	- Counterfeiting - Figuring out - IP piracy	It reduces the area consumption and delay.	High switching activity in the CUT and excessive power consumption
[25]	It proposes a secure scan chain technique using retention loss characteristics of memristors devices for hardware security.	Scan-based attacks are one kind of side-channel attack.	- Suitable for resource-a-constrained implementations. - By comparing the test key, it distinguishes between authorized and unauthorized users. For unauthorized users, it introduces true randomness from degraded memory content to conceal scan responses.	Large arrays increase the overhead
[46]	This study introduces a novel framework called SOTERIA that employs architecture-level improvements and process variation-based authentication signatures to safeguard data in PNoC architectures from snooping attacks.	Snooping attacks	With just modest overheads of up to 10.6% in average latency and up to 13.3% in the energy-delay product (EDP), significantly improve the hardware security in dense-wavelength division-multiplexing (DWDM) based PNoCs.	It has overheads of up to 10.6% in average latency and of up to 13.3% in the EDP compared to the Baseline Photonic NoC (PNoCs).
[26]	It proposes REHAD which has a hardware detection module that can be reconfigured via a trusted software kernel to identify potential attacks and respond to the level of threat. REHAD is a hardware/software co-design for software-based s detection.	Side-channel attacks	- It successfully detects two CSCAs, Flush+Reload and Prime+Probe with different configurations on the reconfigurable detection module. - Dynamic monitoring system (Performed in run time)	Only 5 out of 8 lines of the victim's monitored memory can be detected by the malicious process.
[44]	It proposes a security tag development using femtosecond laser-induced printing of all-dielectric silicon nanoparticles onto polypropylene substrate to protect counterfeiting products.	Counterfeiting attacks	- It is suitable for flexible PUF tag fabrication. - Fast and easy-to-implement method - Promising option to prevent counterfeiting products	N/A
[47]	This work proposes a practical scheme named Dam to mitigate data-oriented attacks using the idea of tagged memory to break data-flow stitching and gadgets dispatcher of generating data-oriented attacks rather than complete Data Flow Integrity (DFI). This method eliminates data-oriented attacks by paying attention to memory access.	Data-oriented attacks	- Average performance cost is 6.48%, lower than DFI which costs 104% - It provides source compatibility and strong security.	It has an average cost overhead of 6.48%.
[27]	It proposes a novel countermeasure called dynamic encoding using a shift register to balance power consumption without precharge at the algorithm level and requires at most 8 random bits at each encryption.	Fault detection side-channel	- It is less expensive. - Smaller area and power overhead. - very well suited to secure lightweight shift, register-based security primitives.	The average power consumption is 1.024 mW.
[28]	It proposes a non-invasive approach that investigates Hardware Intrinsic Attack Detection in IoT (HIADIoT) devices through the power profile of IoT devices in different modes of operation utilizing machine learning algorithms.	Hardware intrinsic: converting channel and power depletion attacks for IoT	It classifies 95.5% of the data point and recognizes potential HI attacks.	It is not applicable to the HI in IC.
[29]	In order to get entropy for a True Random Number Generator (TRNG) and Physical Unclonable Function (PUF), dynamic Threshold Switching (TS) variations and static leakage current mismatch are used in this proposed method. This is a unique approach to combining TRNG and PUF in a 4-layer 3D threshold switching (TS) NbOx.	Immunity against power noise attack (frequency range up to 400MHz, and amplitude range up to 600mV)	- Hardware solution for low-cost secure integrated systems. - TRNG presents the ultra-low energy consumption of 0.75 pJ/bit at 2.0 V.	It has a 1.66% of native bit error rate.

TABLE 2. (Continued.) A comparison between the existing hardware security methods.

[30]	It proposes machine learning to augment the multiphysics simulation framework and efficiently generate time-domain EM traces at any virtual probe above the surface of the silicon substrate by modeling cell-level power, chip logic functionality, and layout geometry. In order to predict the most vulnerable leakage location, ML-based automatic POI (point-of-interest) identification algorithm was also proposed in this work.	Near-field Electromagnetic (EM) side-channel leakage	- Identifying the leaking points that are most susceptible. - Compared to a typical correlation-based side-channel simulation method, this approach is 10-100x faster.	130nm AES128 test chip is used for validation of the method.
[31]	In a wet-etching-based 3D package structure, two dies are embedded in the substrate trench, and one die flips down and connects with the two dies via the redistribution layer. The embedded die is surrounded by metal wire, which constructs a metal shielding network and can effectively prevent electromagnetic radiation leakage from the die. The metal shielding network can also act as an invasive attack detection circuit.	- EM radiation (side-channel) - Reverse engineering attempts	- Low cost - Smaller form factor and higher performance	- Testing flow of 3D is more complex compared to 2D - An adversary could gather some of the system's thermal data.
[32]	CMOS detector circuit combined with BBM Cu meander is proposed to defend the cryptographic circuits against physical security attacks from vulnerable backside Si substrates. The meander wire pattern using backside buried materials (BBM) on the Si backside uses the idea of disconnection of the meander and warns the malicious attempts on the Si substrate.	- Backside physical security attack in flip-chip packaging. - Passive side-channel attacks and active laser fault injection	It warns the malicious attempts to expose a vulnerable Si substrate.	-0.0025% size overhead
[48]	A hardware architecture is introduced and demonstrated showing the efficiency of traffic modeling and the ability to detect two different forms of hardware attacks on the I2C a most popular serial communication bus.	Vulnerabilities focus on I2C busses	- Self-monitoring security solution that will assess an abnormal behavior - Real-time and low-cost - It is compatible to be embedded with small IoT devices.	Results might differ for other communication buses rather than I2C busses.
[34]	This study proposes improvements to the MuonTrap hierarchy called Speculative yet Secure Prefetching (SpecPref), which permits prefetching both on commit and speculatively.	Hardware prefetcher attack	It improves the performance slow down to 1.17% compared to 20.46% in MuonTrap [81].	It has marginal changes to the cache hierarchy.
[39]	It proposes processor extensions for hardware instruction replay against fault injection attacks.	Fault injection attacks	- Faster execution time 75% fault detected	It has +30% core area overhead.
[54]	It proposes a Deceptive Multiplexer (D-MUX) based logic locking scheme which is resilient against structure-exploiting machine learning attacks.	Learning-based attacks	It is efficient in protecting against SnapShot.	- Key-related information may only be disclosed by Logic locking at the topological level. - Structural leakage may be present in the design. - D-MUX may also leak information, which is beyond human-eye observation. - Since the power overhead is calculated using DC, it increases as the area overhead does.
[33]	Thread-level monitoring of micro-architectural event counters at the run time was proposed to protect against cache side-channel attacks. Either manually or automatically sensitive thread of the user application needs to be Marked first. Micro-architectural event counts will be observed and compared with the predetermined threshold value, and the attack will be detected if the count value exceeds the threshold value.	Cache side-channel attacks	- It has greater precision and dependability than system-level monitoring. - Fine-grained approach	It is limited to low-frequency applications.

TABLE 2. (Continued.) A comparison between the existing hardware security methods.

[77]	It proposes a PRIME+ABORT detection method that records selected hardware event counts using the hardware counter during AES decryption and the PRIME+ABORT attack. Count for the selected event was performed every second. AES decryption was run 11200 times, and PRIME and ABORT attack was run 8500 times.	• Cash side-channel attacks	• 99.5% success rate during 30 minutes of execution time	• The detector requires careful modification on which it will run to acquire acceptable detection performance.
[50]	The proposed architecture uses a CoAP accelerator that cooperates with the device's CPU to alleviate the IoT SoC's CPU burden of handling incoming and outgoing CoAP messages. When the CoAP is activated, it will handle messages without CPU interaction. As a result, the CPU will be free to execute helpful work. A prototype of the proposed architecture was also implemented in SoC FPGA, and an evaluation was also carried out to see if the objective was satisfied.	• Distributed Denial of Service (DDoS)	• It provides resiliency in IOT devices against DDoS attack.	• It does not provide resiliency against all sorts.
[45]	THz range stays between the microwave and infrared range. THz bridges the electronic and optical portions of the spectrum, but it suffers from poor spatial resolution. A work by [45] an advanced metrology method to detect authentic vs counterfeit IC. The proposed method uses THz point spread function to improve spatial resolution. This high-fidelity model compares the digital structural and spectral properties in unsupervised deep learning (DL).	• DDoS • IC counterfeit	• It has the ability of automated counterfeit detection. • Minimal to no manual labor is required.	• The overall spatial and spectral resolution of THz needs to be improved.
[51]	A hardware-integrated system for determining the memory capacity bounds was developed to provide a simple countermeasure to buffer overflow attacks.	• Buffer overflow attacks	• It allows both passcode verification and checking the buffer memory's boundary limits. • It allows usage of a minimalistic number of logic gates as an encryption method.	• Cost is not discussed

4) MAN-IN-THE-MIDDLE ATTACK ON PCIE BUS FOR SMART DATA REPLAY

The architecture of a Stratix 5 FPGA-based platform is proposed by Khelif et al. [85] to conduct a hardware Man-in-the-Middle (MitM) attack on a computer's PCIe 3.0 bus and emulate the behavior of an SoC. This MitM can log, edit, and copy data into a shadow memory so that it can be replayed to the host. The Man-in-the-Middle attack, expressed as MitM, consists of inserting a third device into an end-to-end communication. Between the two peripherals, this device behaves as an invisible router. By recording and interfering with the transmitted data in real-time, the attacker in this position has the ability to undermine the communication's confidentiality and integrity. Hardware MitM is a special attack that provides direct access to data and offers various attack vector possibilities, including data replay, traffic analysis, and fault injection. It is also one of the least exploited attacks in the state of the art but is also one of the most promising. All the information kept in memory, including the counter for password attempts, is encrypted, but a hardware MitM attack may still damage it.

5) HARDWARE TROJAN ATTACKS ON PROCESSOR

The main idea is to use conditionally triggered hardware Trojan Attacks on processor cores based on time information as the trigger conditions. Research by Kuo et al. [86] present two types of Hardware Trojan designs depending on the time information as the trigger conditions, one using real-world time and the other using the time passed since power-on. A programmable scheme was also designed in this study to set a trigger condition by controlling the scan chain of a system. This work also shows how to insert these Trojan designs in an open-source processor core (RISC-V processor). This type of Trojans typically comprises two parts: the trigger and the payload. The trigger part monitors the internal or external signals of the host circuit. When some specific, rare condition is satisfied, the payload part starts to attack the host design. The trigger part can be implemented using combinational, sequential, or both logic. Suggested method has low area and power overhead as they have used existing circuit components along with a counter to design the Trojan.

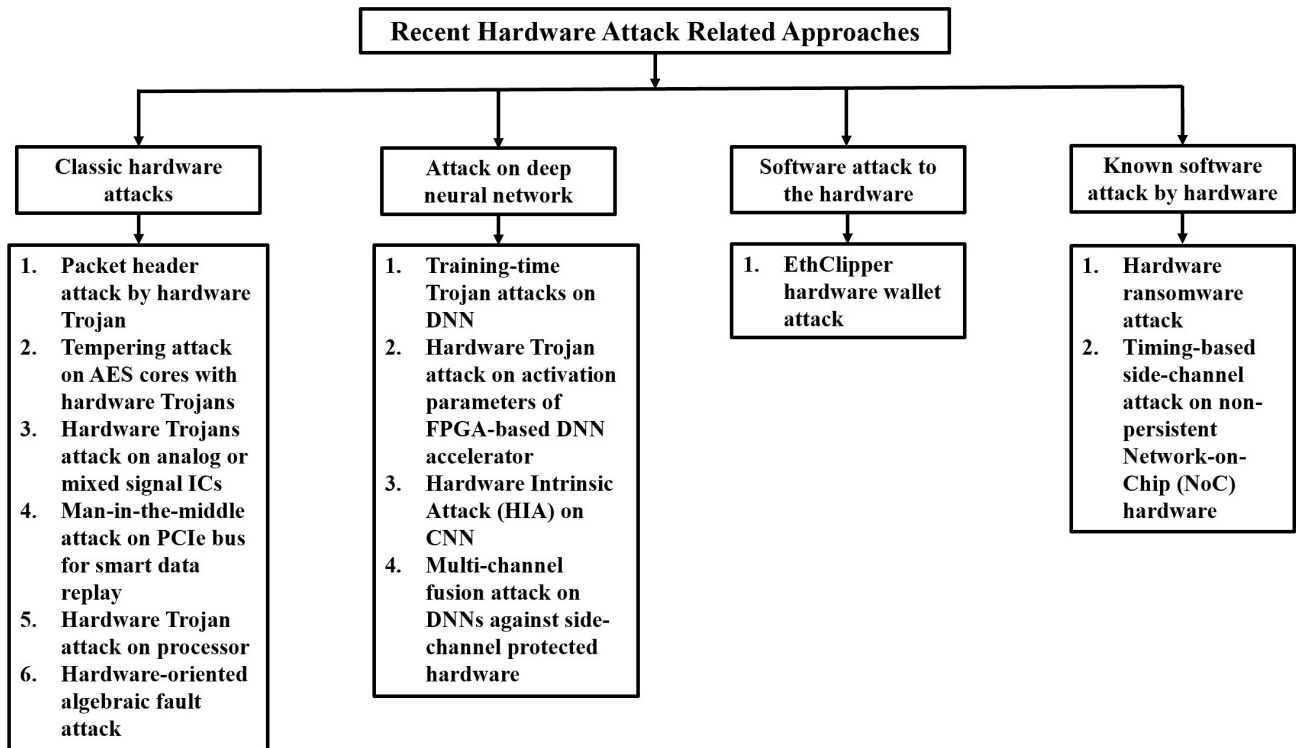


FIGURE 3. Hardware attack approaches.

6) HARDWARE-ORIENTED ALGEBRAIC FAULT ATTACK

Gay et al. [87] propose a framework that can construct fault attacks automatically on hardware-oriented structural cipher descriptions which can quickly evaluate any cipher implementations, including any optimizations. But the output of the framework is a set of algebraic equations which is then fed to a SAT solver. It takes two primary inputs, namely the circuit model and the fault descriptions. It takes two primary inputs, the circuit model and the fault description. The fault description mainly consists of three parts; 1) The fault round (e.g., in which round or at which operation the fault is injected), 2) the fault location (e.g., the byte or nibble affected by the fault), and 3) the number of faults. The proposed framework is the first Algebraic Fault Attack (AFA) tool to examine full-scale AES, LED-64, and PRESENT with only hardware-oriented structural cipher descriptions.

B. ATTACK ON DEEP NEURAL NETWORK

1) TRAINING-TIME TROJAN ATTACKS (DEEP NEURAL NETWORK (DNN) TROJAN ATTACKS)

Model Trojaning attacks, in which adversaries introduce malicious backdoors into ML models, are one of the most severe security vulnerabilities in DNNs. DNN Trojan attacks can cause a victim model to purposefully misclassify a subset of inputs via a Trojan trigger, with little to no impact on the inference of actual inputs. Previous training time attack literature typically can be divided into two groups: i) A Data poisoning attack [88] is when victim models are

trained using maliciously constructed poisoned data samples, and ii) A Re-training-based Trojan attack is when adversaries can arbitrarily alter model parameters through local training [89]. One recently reported Hardware-based fault attack is Rowhammer [90], which demonstrated practical concerns about the integrity of computing systems. This attack is more severe because fault attack vectors allow internal tampering of a target system even when the data and software stack are configured correctly and controlled by legitimate users. The recent study also shows Inference time adversarial attacks, which inject bit flips into DNN models stored in memory to compromise model classification behaviors [91]. Moreover, recent work by Cai et al. [92] provide hardware-induced memory faults as a means of training-time Trojan attacks. During machine learning training, the proposed attack methodology perturbs the feature maps of a victim model via bit flips and finds the Trojan trigger pattern during inference time. This framework provides a high attack success rate on average of 99.98% on misclassification.

2) HARDWARE TROJAN ATTACK ON ACTIVATION PARAMETERS OF FPGA-BASED DNN ACCELERATORS

The work by Mukherjee and Chakraborty [93] propose a noble hardware Trojan-based attack on the internal activation parameters of two trained DNN hardware accelerators. The trained DNN hardware accelerators are intended for object identification and image classification and are targeted toward the FPGA platform. When successfully triggered, the

TABLE 3. Summary of hardware attack related approaches.

Ref.	Attack Name	Description of the attack	Strength	Weakness
[92]	Training-time Trojan attacks (DNN trojan attacks)	It offers training-time Trojan attacks through hardware-induced faults in memory. The proposed attack framework perturbs a victim model's feature maps via bit flips during machine learning training and identifies the Trojan trigger pattern during inference time. This framework provides a high attack success rate (99.98% on average) misclassification.	High attack success rate 99.98%	Efficacy needs to be tested in a real system.
[93]	Hardware Trojan attack on activation parameters of FPGA-Based DNN accelerators	The suggested HT consists of a glitch-generating circuitry that, when successfully activated, alters the activation parameters of the DNN in such an accelerator. The trained DNN hardware accelerators are intended for object identification and image classification and are targeted toward the FPGA platform.	- Resource footprint of the proposed HT is very low. - Noise in input data cannot be used as a countermeasure. - It does not have a noticeable impact by the random cropping and rotations $[-5^\circ$ to $+5^\circ]$ of the input images. - Claimed as the most effective.	As existing work used different data sets, different neural architecture, and proposed HT against the other attack, comparing resource overhead with the existing work was challenging. The results table in this article only shows the relative effectiveness of the proposed HT.
[96]	EthClipper is a hardware wallet clipboard meddling attack with address verification evasion.	In EthClipper, the attacker deploys a distributed system called ClipperCloud, which mines and stores billions of Ethereum accounts. When the malware finds an Ethereum address in the clipboard, it asks ClipperCloud to locate the mined account with the most visual similarities to the address in the clipboard among the accounts created. As a result, the victim's confirmation bias could well be activated by the visual similarity between the address on the screen and the expected address, leading to the approval of the fraudulent transaction.	- The attack can be carried out with a limited budget on retail equipment. - Fool its owner into confirming a malicious transaction.	To increase the likelihood of discovering a suitable fake location by 50%, an address must match 25% of the digits.
[97]	Hardware ransomware attacks	This is a specialized form of hardware Trojan, therefore, termed hardware ransomware which mounts a ransomware attack in hardware by combining cryptographic hardware and a key-generation method.	- Hardware ransomware attack is possible. - Claimed as the first work of hardware ransomware attack.	It shows the feasibility in 65 nm CMOS, but the result could be different in 3nm technology.
[99]	Timing-based side-channel attack on non-persistent NoC hardware,	In order to reduce noise and increase predictability, ConNOC suggests a novel attack architecture that better takes use of interference at NoC hardware.	- ConNOC exhibits an 11x improvement over baseline on a real CPU. - The evaluation of covert communication and information leakage attacks demonstrates 100 percent accuracy when employing five replays to leak information.	It has an average latency variance of 7.3 cycles.
[82]	Packet header attack by hardware Trojan	It models an HT that can alter the destination address field of specific NoC packets and is mounted on the input buffers of NoC routers.	It can severely affect the miss penalty of L1 caches and even stop an application by blocking an instruction.	It only applies to situations where the cores perform low injection rate programs.
[94]	Hardware Intrinsic Attack (HIA) on CNN	HIA is inserted inside mathematical operations of specific CNN layers, propagating erroneous operations in all the succeeding CNN layers and resulting in misclassification. The attack propagates incorrect feature mappings by rearranging the weight matrices of convolutional layers. Five alternative attack scenarios are developed for each CNN layer and assessed in terms of the rate of triggering and overhead resources compared to the initial implementation.	Additional latency is minimal in all attack scenarios (0:61%), and there is also a minimal increase in DSP, LUT, and FF (2:36% in each case).	There are slight variations in BRAM that are evident in three of the five tested scenarios.

TABLE 3. (Continued.) Summary of hardware attack related approaches.

[95]	Multi-Channel fusion attack framework based on DNNs against side-channel protected hardware	The proposed method extracts location-dependent leakages that capture the activity of specific parts of the hardware design using different decoupling capacitors on a modern system-on-chip (SoC) using 16nm fabrication technology of the power supply. A multi-input DNN is then used to analyze the collected data. This network has been trained to combine share-related data to forecast outcomes for the processed masked data.	No knowledge of the mask values is required for a successful attack in the proposed method, hence it works with more realistic adversary assumptions.	The implementation displays nonfirst-order usual leakage due to coupling effects.
[83]	Tampering attack on AES core with hardware Trojans	The proposed method depends on when an untrusted foundry inserts a hardware Trojan in the netlist. Proposed trojan activation can modify the input data of a particular round by preventing key-dependent computation of the previous round. Then the proposed sequential hardware Trojan helps the attacker to collect this modified data at the input of an internal round.	Since the attacker only knows the trigger condition generated from the input plaintext and applies it repeatedly, it is very challenging to identify such Trojans using logic testing.	An overhead is caused by the payload because it needs 128 OR gates.
[84]	Hardware Trojan attacks on analog/mixed-signal ICs & snooping attacks	The proposed HT assault scenario in which the Trojan does not reside inside the analog circuit of the victim. However, it is located on the independent digital circuit on the same die where it was triggered. Its payload is only applied to the analog circuit after being transmitted via the common test infrastructure and the analog circuit's test interface. This HT attack uses the dense digital circuit to conceal its footprint successfully, making it impossible to detect in the analog domain. To increase the difficulty of detection and prevention, the HT is concealed and activated inside a digital IP block and the SoC's test infrastructure, and its payload is transferred to the analog IP block by taking advantage of the typical Test Access Mechanism (TAM), which creates a daisy-chain network connecting all of the IP blocks in an SoC and applying it to the analog IP block via the Built-in Self-Test (BIST) interface.	This study claimed, "this HT attack cannot be detected or prevented in the analog domain".	One defense strategy is to encrypt the scan route within the digital IP, so that, the intent HT triggers a signal from the digital IP when it passes through the scan path, compromising the intent payload.
[85]	Man-in-the-middle attack on PCIe bus for smart data replay.	The man-in-the-middle attack, expressed as MitM, consists of inserting a third device into an end-to-end communication. Between the two peripherals, this device behaves as an invisible roofer. By recording and interfering with the transmitted data in real-time, the attacker in this position has the ability to undermine the communication's confidentiality and integrity. Hardware MitM is a special attack that provides direct access to data and offers various attack vector possibilities, including data replay, traffic analysis, and fault injection.	All the information kept in memory, including the counter for password attempts, is encrypted, but a hardware MitM attack may still damage it.	It is difficult to create an architecture constrained protocols in terms of throughput, response time, and frequency.
[86]	Conditionally triggered hardware Trojan attacks on processor cores based on time information as the trigger conditions.	This type of Trojans typically comprises of two parts: the trigger and the payload. The trigger part monitors the internal or external signals of the host circuit. When some specific, rare condition is satisfied, the payload part starts to attack the host design. The trigger part can be implemented using combinational, sequential, or both logic.	Low area and power overhead as they have used existing circuit components along with a counter to design the Trojan.	It has delay overhead

glitch-generating circuitry in the proposed HT modifies the activation parameters of the DNN in such an accelerator.

3) HARDWARE INTRINSIC ATTACK (HIA) ON CONVOLUTIONAL NEURAL NETWORK (CNN)

Odetola and Hasan [94] present a noble Hardware intrinsic attack (HIA). The proposed HIA is inserted inside mathematical operations of specific CNN layers, propagating

erroneous operations in all the succeeding CNN layers and resulting in misclassification. The attack propagates incorrect feature mappings by rearranging the weight matrices of convolutional layers. This work also shows five alternative attack scenarios that are developed for each CNN layer and assessed in terms of the rate of triggering and overhead resources compared to the initial implementation. Additional latency is minimal in all attack scenarios (0:61%), and there is

also a minimal increase in DSP, LUT, and FF (2:36% in each case). Only very slight variations in Block RAM (BRAM) are evident in three of the five tested scenarios. The parameters of the model are not altered to carry out this attack. Attacks are entirely random and non-periodic, making them hard to identify.

4) MULTI-CHANNEL FUSION ATTACK FRAMEWORK BASED ON DNNs AGAINST SIDE-CHANNEL PROTECTED HARDWARE
Multi-channel fusion framework based on DNNs is proposed by Hettwer et al. [95]. The proposed method extracts location-dependent leakages that capture the activity of specific parts of the hardware design using different decoupling capacitors on a modern SoC using 16nm fabrication technology of the power supply. A multi-input DNN is then used to analyze the collected data. This network has been taught to aggregate share-related data in order to forecast the results of processed masked data. This method works with more realistic adversary assumptions as no mask values need to be known for a successful attack.

C. SOFTWARE ATTACK TO THE HARDWARE

1) ETHCLIPPER HARDWARE WALLET ATTACK

Ivanov and Yan [96] propose an attack named EthClipper, which goes after owners of hardware wallets on the Ethereum platform. EthClipper is a hardware wallet clipboard meddling attack with address verification evasion to choose the address with the highest visual similarity to the original one, the EthClipper malware searches a distributed database of pre-mined accounts. EthClipper, adds a social engineering component to the existing clipboard substitution technique. A distributed system named ClipperCloud is used by the hacker in EthClipper to mine and store billions of Ethereum accounts. When the malware discovers an Ethereum address in the clipboard, it instructs ClipperCloud to find the mined account among the accounts generated with the greatest visual resemblance to the address in the clipboard. Because the address on the screen and the expected address appear to be visually similar, the victim's confirmation bias may be triggered, resulting in the victim's approval of the fraudulent transaction. To carry out this attack only a limited budget is required and can cause malicious transactions.

D. KNOWN SOFTWARE ATTACK BY HARDWARE

1) HARDWARE RANSOMWARE ATTACK

Almeida et al. [97] propose a Ransomware attack which was previously existed in the software domain only. Ransomware is a category of malicious software (sometimes known as malware) intended to encrypt or restrict user access to data [98].

Typically, a ransomware attack is triggered by the user or a timing event, and all user data is encrypted. A specialized hardware Trojan attack called hardware Ransomware attack is proposed in this study by combining cryptographic hardware and a key generation scheme. This study examines the viability of Hardware Ransomware attacks and offered com-

prehensive instructions for mapping the design to multiple hardware platforms, including FPGA and ASIC, along with a silicon implementation in 65nm CMOS. It is claimed in this work that Ransomware attack is possible on the hardware.

2) TIMING BASED SIDE CHANNEL ATTACK ON NON-PERSISTENT NETWORK-ON-CHIP (NOC) HARDWARE

Ali and Khan [99] provide a Timing-based side-channel attack on non-persistent NoC hardware, and demonstrates its practicality on a real multicore machine. The provided novel attack setup better exploits interference at NoC hardware to make it less noisy and more predictable. The proposed method, named ConNOC, exhibits an 11x improvement over baseline on a real CPU and shows 100% accuracy in covert communication and information leakage when employing five replays to leak information.

VI. CONCLUSION

This paper investigates state-of-the-art hardware security trends. We have discussed the available framework and methods proposed by the researcher, along with some challenges for tackling different domains of the hardware security field. It is apparent from the literature that researchers have recently paid close attention to language-based hardware security solutions. However, attacks in Deep Neural Networks (DNN) also recently have received huge attention from researchers. It is found that most of the challenges faced by the researcher are balancing security and resource usage, especially for lightweight/battery-constrained devices. Moreover, this paper also presents the benefits and drawbacks of the current hardware security methods. In addition to that, this work also presents the most common hardware attack techniques and indicates the strengths and weaknesses of each one. However, with new technology, new hardware security vulnerabilities will emerge. So, understanding challenges and existing security methods applicable to particular attacks, along with the advantages and drawbacks of the method, will help the practitioner to choose the appropriate approach to address emerging hardware attacks in this challenging security field.

REFERENCES

- [1] W. Hu, C.-H. Chang, A. Sengupta, S. Bhunia, R. Kastner, and H. Li, "An overview of hardware security and trust: Threats, countermeasures, and design tools," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 40, no. 6, pp. 1010–1038, Jun. 2021.
- [2] B. Halak and M. S. Mispan, "An overview of machine learning applications in hardware security," in *Proc. IEEE Int. Conf. Design Test Integr. Micro Nano-Syst. (DTS)*, Jun. 2022, pp. 1–6.
- [3] X. Guo, R. G. Dutta, J. He, M. M. Tehranipoor, and Y. Jin, "QIF-verilog: Quantitative information-flow based hardware description languages for pre-silicon security assessment," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, May 2019, pp. 91–100.
- [4] S. Skorobogatov. (2012). *Hardware Assurance and Its Importance to National Security*. [Online]. Available: <http://www.cl.cam.ac.uk/sps32/secnews.html>
- [5] J. Grand, J. Appelbaum, and C. Tarnovsky, "'Smart' parking meter implementations. Globalism, and you," *Comput. Sci., Tech. Rep.* 221001481, 2009. [Online]. Available: <https://research.tue.nl/en/publications/smart-parking-meterimplementations-globalism-and-you>

- [6] M. Nagata, T. Miki, and N. Miura, "Physical attack protection techniques for IC chip level hardware security," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 30, no. 1, pp. 5–14, Jan. 2022.
- [7] H. Wei, M. Yan, and M. Onabajo, "Noise reduction via chopper stabilization of fully differential temperature sensors for hardware security applications," in *Proc. IEEE 14th Dallas Circuits Syst. Conf. (DCAS)*, Nov. 2020, pp. 1–5.
- [8] K. Khalil, K. Elgazzar, and M. Bayoumi, "A comparative analysis on resource discovery protocols for the Internet of Things," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2018, pp. 1–7.
- [9] I. U. Din, M. Guizani, S. Hassan, B.-S. Kim, M. K. Khan, M. Atiquzzaman, and S. H. Ahmed, "The Internet of Things: A review of enabled technologies and future challenges," *IEEE Access*, vol. 7, pp. 7606–7640, 2019.
- [10] K. Khalil, K. Elgazzar, M. Seliem, and M. Bayoumi, "Resource discovery techniques in the Internet of Things: A review," *Internet Things*, vol. 12, Dec. 2020, Art. no. 100293.
- [11] *Internet of Things (IoT) Market Size, Share & COVID-19 Impact Analysis, by Component (Platform, Solution & Services), by End-Use Industry (BFSI, Retail, Government, Healthcare, Manufacturing, Agriculture, Sustainable Energy, Transportation, it & Telecom, Others), and Regional Forecast, 2021–2028*, Fortune Bus. Insights, Pune, India, 2021.
- [12] K. Khalil, A. Abdelgawad, and M. Bayoumi, "Intelligent resource discovery approach for the Internet of Things," in *Proc. IEEE 7th World Forum Internet Things (WF-IoT)*, Jun. 2021, pp. 264–269.
- [13] Y. Ali, S. W. Shah, and W. A. Khan, "Security at the Internet of Things," in *Machine Tools*. Boca Raton, FL, USA: CRC Press, 2023, pp. 31–48.
- [14] K. Khalil, K. Elgazzar, A. Abdelgawad, and M. Bayoumi, "A security approach for CoAP-based Internet of Things resource discovery," in *Proc. IEEE 6th World Forum Internet Things (WF-IoT)*, Jun. 2020, pp. 1–6.
- [15] W. Chen, R. Huo, C. Sun, S. Zeng, S. Wang, and T. Huang, "SCRT: A secure and efficient state-channel-based resource trading scheme for Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 11, pp. 10038–10051, Jun. 2023.
- [16] S. Sidhu, B. J. Mohd, and T. Hayajneh, "Hardware security in IoT devices with emphasis on hardware Trojans," *J. Sensor Actuator Netw.*, vol. 8, no. 3, p. 42, Aug. 2019.
- [17] M. Rostami, F. Koushanfar, and R. Karri, "A primer on hardware security: Models, methods, and metrics," *Proc. IEEE*, vol. 102, no. 8, pp. 1283–1295, Aug. 2014.
- [18] J. G. Rajan, and R. S. Ganesh, "Hardware based data security techniques in IoT: A review," in *Proc. 3rd Int. Conf. Smart Electron. Commun. (ICOSEC)*, Oct. 2022, pp. 408–413.
- [19] A. Sharma, G. O. Dyrkolbotn, L. Øverlier, A. J. Waltoft-Olsen, K. Franke, and S. Katsikas, "A state-of-the-art reverse engineering approach for combating hardware security vulnerabilities at the system and PCB level in IoT devices," in *Proc. IEEE Phys. Assurance Inspection Electron. (PAINE)*, Oct. 2022, pp. 1–7.
- [20] D. Dobrykh, D. Filonov, A. Slobozhanyuk, and P. Ginzburg, "Hardware RFID security for preventing far-field attacks," *IEEE Trans. Antennas Propag.*, vol. 70, no. 3, pp. 2199–2204, Mar. 2022.
- [21] B. Kumar, A. K. Jaiswal, V. S. Vineesh, and R. Shinde, "Analyzing hardware security properties of processors through model checking," in *Proc. 33rd Int. Conf. VLSI Design 19th Int. Conf. Embedded Syst. (VLSID)*, Jan. 2020, pp. 107–112.
- [22] M. Montoya, T. Hiscock, S. Bacles-Min, A. Molnos, and J. Fournier, "Adaptive masking: A dynamic trade-off between energy consumption and hardware security," in *Proc. IEEE 37th Int. Conf. Comput. Design (ICCD)*, Nov. 2019, pp. 559–566.
- [23] D. Divyanshu, R. Kumar, D. Khan, S. Amara, and Y. Massoud, "Logic locking using emerging 2T/3T magnetic tunnel junctions for hardware security," *IEEE Access*, vol. 10, pp. 102386–102395, 2022.
- [24] W. Ho, A. A. Pammu, K. Z. L. Ne, K.-S. Chong, and B.-H. Gwee, "Reconfigurable routing paths as noise generators using NoC platform for hardware security applications," in *Proc. 32nd IEEE Int. Syst.-on-Chip Conf. (SOCC)*, Sep. 2019, pp. 86–91.
- [25] Y. Gong, F. Qian, and L. Wang, "Design for test and hardware security utilizing retention loss of memristors," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 27, no. 11, pp. 2536–2547, Nov. 2019.
- [26] Y. Mao, V. Migliore, and V. Nicomette, "REHAD: Using low-frequency reconfigurable hardware for cache side-channel attacks detection," in *Proc. IEEE Eur. Symp. Secur. Privacy Workshops (EuroSPW)*, Sep. 2020, pp. 704–709.
- [27] M. Montoya, S. Bacles-Min, A. Molnos, and J. J. A. Fournier, "Dynamic encoding, a lightweight combined countermeasure against hardware attacks," in *Proc. 23rd Euromicro Conf. Digit. Syst. Design (DSD)*, Aug. 2020, pp. 185–192.
- [28] H. Mohammed, T. A. Odetola, S. R. Hasan, S. Stissi, I. Garlin, and F. Awwad, "(HIADIoT): Hardware intrinsic attack detection in Internet of Things; leveraging power profiling," in *Proc. IEEE 62nd Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Aug. 2019, pp. 852–855.
- [29] Q. Ding, H. Jiang, J. Li, C. Liu, J. Yu, P. Chen, Y. Zhao, Y. Ding, T. Gong, J. Yang, Q. Luo, Q. Liu, H. Lv, and M. Liu, "Unified 0.75 pJ/bit TRNG and attack resilient 2F2/bit PUF for robust hardware security solutions with 4-layer stacking 3D NbO_x threshold switching array," in *IEDM Tech. Dig.*, Dec. 2021, pp. 39.2.1–39.2.4.
- [30] L. Lin, D. Zhu, J. Wen, H. Chen, Y. Lu, N. Chang, C. Chow, H. Shrivastav, C. Chen, K. Monta, and M. Nagata, "Multiphysics simulation of EM side-channels from silicon backside with ML-based auto-POI identification," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, Dec. 2021, pp. 270–280.
- [31] C. Zhu, Y. Yan, P. Guo, and J. Li, "Leveraging 3D packaging technology to enhance integrated circuits security and reliability," in *Proc. 19th Int. Conf. Electron. Packag. Technol. (ICEPT)*, Aug. 2018, pp. 766–769.
- [32] T. Miki, M. Nagata, H. Sonoda, N. Miura, T. Okidono, Y. Araga, N. Watanabe, H. Shimamoto, and K. Kikuchi, "A Si-backside protection circuits against physical security attacks on flip-chip devices," in *Proc. IEEE Asian Solid-State Circuits Conf. (A-SSCC)*, Nov. 2019, pp. 25–28.
- [33] P. P. Bhade and S. Sinha, "Detection of cache side channel attacks using thread level monitoring of hardware performance counters," in *Proc. IEEE 14th Int. Symp. Embedded Multicore/Many-Core Syst.-on-Chip (MCSoc)*, Dec. 2021, pp. 210–217.
- [34] T. Solanki and B. Panda, "SpecPref: High performing speculative attacks resilient hardware prefetchers," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, Jun. 2022, pp. 57–60.
- [35] M. Eslami, T. Ghasempouri, and S. Pagliarini, "Reusing verification assertions as security checkers for hardware trojan detection," 2022, *arXiv:2201.01130*.
- [36] L. M. Reimann, L. Hanel, D. Sisejkovic, F. Merchant, and R. Leupers, "QFlow: Quantitative information flow for security-aware hardware design in verilog," in *Proc. IEEE 39th Int. Conf. Comput. Design (ICCD)*, Oct. 2021, pp. 603–607.
- [37] S. Yu, R. Yasaei, Q. Zhou, T. Nguyen, and M. A. Al Faruque, "HW2VEC: A graph learning tool for automating hardware security," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, Dec. 2021, pp. 13–23.
- [38] O. Arias, Z. Liu, X. Guo, Y. Jin, and S. Wang, "RTSEC: Automated RTL code augmentation for hardware security enhancement," in *Proc. Design, Autom. Test Eur. Conf. Exhib. (DATE)*, Mar. 2022, pp. 596–599.
- [39] N. A. Manssour, V. Lapotre, G. Gogniat, and A. Tisserand, "Processor extensions for hardware instruction replay against fault injection attacks," in *Proc. 25th Int. Symp. Design Diagnostics Electron. Circuits Syst. (DDECS)*, 2022, pp. 26–31.
- [40] R. Naveenkumar, N. M. Sivamangai, A. Napoleon, A. Puviarasu, and G. Saranya, "Preventive measure of SAT attack by integrating anti-SAT on locked circuit for improving hardware security," in *Proc. 7th Int. Conf. Commun. Electron. Syst. (ICCES)*, Jun. 2022, pp. 756–760.
- [41] S. Patnaik, N. Rangarajan, J. Knechtel, O. Sinanoglu, and S. Rakheja, "Spin-orbit torque devices for hardware security: From deterministic to probabilistic regime," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 39, no. 8, pp. 1591–1606, Aug. 2020.
- [42] T. Thangam, G. Gayathri, and T. Madhubala, "A novel logic locking technique for hardware security," in *Proc. IEEE Int. Conf. Electr. Instrum. Commun. Eng. (ICEICE)*, Apr. 2017, pp. 1–7.
- [43] S. Dutta, B. Grisafe, C. Frentzel, Z. Enciso, M. S. Jose, J. Smith, K. Ni, S. Joshi, and S. Datta, "Experimental demonstration of gate-level logic camouflaging and run-time reconfigurability using ferroelectric FET for hardware security," *IEEE Trans. Electron Devices*, vol. 68, no. 2, pp. 516–522, Feb. 2021.
- [44] P. Kustov, E. Petrova, M. Sandomirskii, and D. Zuev, "All-dielectric silicon nanoparticles on flexible substrate for anticounterfeiting labels," in *Proc. 16th Int. Congr. Artif. Mater. Novel Wave Phenomena (Metamaterials)*, Sep. 2022, pp. 244–246.
- [45] J. True, C. Xi, N. Jessurun, K. Ahi, M. Tehranipoor, and N. Asadizanjani, "TeraHertz based machine learning approach to integrated circuit assurance," in *Proc. IEEE 71st Electron. Compon. Technol. Conf. (ECTC)*, Jun. 2021, pp. 2235–2245.

- [46] S. V. R. Chittamuru, I. G. Thakkar, V. Bhat, and S. Pasricha, "SOTERIA: Exploiting process variations to enhance hardware security with photonic NoC architectures," in *Proc. 55th ACM/ESDA/IEEE Design Autom. Conf. (DAC)*, Jun. 2018, pp. 1–6.
- [47] M. Ma, L. Chen, and G. Shi, "Dam: A practical scheme to mitigate data-oriented attacks with tagged memory based on hardware," in *Proc. 26th Asia-Pacific Softw. Eng. Conf. (APSEC)*, Dec. 2019, pp. 204–211.
- [48] J. Lorandel, M. A. Khelif, and O. Romain, "A low-cost hardware attack detection solution for IoT devices," in *Proc. IEEE 31st Int. Symp. Ind. Electron. (ISIE)*, Jun. 2022, pp. 674–679.
- [49] G. Gogniat, T. Wolf, W. Burleson, J.-P. Diguët, L. Bossuet, and R. Vaslin, "Reconfigurable hardware for high-security/ high-performance embedded systems: The SAFES perspective," *IEEE Trans. Very Large Scale Integr. (VLSI) Syst.*, vol. 16, no. 2, pp. 144–155, Feb. 2008.
- [50] L. R. B. Brasilino and M. Swamy, "Mitigating DDoS flooding attacks against IoT using custom hardware modules," in *Proc. 6th Int. Conf. Internet Things: Syst., Manage. Secur. (IOTSMS)*, Oct. 2019, pp. 58–64.
- [51] S. Sayeshwari and E. Prabhu, "A simple countermeasure to mitigate buffer overflow attack using minimalistic hardware-integrated software simulation for FPGA," in *Proc. IEEE Int. Conf. Electron., Comput. Commun. Technol. (CONECT)*, Jul. 2022, pp. 1–4.
- [52] C. Bernard, W. Bryant, R. Becker, and J. Di, "Design of asynchronous polymorphic logic gates for hardware security," in *Proc. IEEE High Perform. Extreme Comput. Conf. (HPEC)*, Sep. 2021, pp. 1–5.
- [53] G. Dessouky, S. Zeitouni, A. Ibrahim, L. Davi, and A. Sadeghi, "CHASE: A configurable hardware-assisted security extension for real-time systems," in *Proc. IEEE/ACM Int. Conf. Computer-Aided Design (ICCAD)*, Nov. 2019, pp. 1–8.
- [54] D. Sisejkovic, F. Merchant, L. M. Reimann, and R. Leupers, "Deceptive logic locking for hardware integrity protection against machine learning attacks," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 41, no. 6, pp. 1716–1729, Jun. 2022.
- [55] R. Karri, J. Rajendran, K. Rosenfeld, and M. Tehranipoor, "Trustworthy hardware: Identifying and classifying hardware trojans," *Computer*, vol. 43, no. 10, pp. 39–46, Oct. 2010.
- [56] M. Tehranipoor and F. Koushanfar, "A survey of hardware trojan taxonomy and detection," *IEEE Design Test Comput.*, vol. 27, no. 1, pp. 10–25, Jan. 2010.
- [57] C. Sturton, M. Hicks, D. Wagner, and S. T. King, "Defeating UCI: Building stealthy and malicious hardware," in *Proc. IEEE Symp. Secur. Privacy*, May 2011, pp. 64–77.
- [58] M. Hicks, M. Finnicum, S. T. King, M. M. K. Martin, and J. M. Smith, "Overcoming an untrusted computing base: Detecting and removing malicious hardware automatically," in *Proc. IEEE Symp. Secur. Privacy*, May 2010, pp. 159–172.
- [59] J. A. Roy, F. Koushanfar, and I. L. Markov, "EPIC: Ending piracy of integrated circuits," in *Proc. Conf. Design, Autom. Test Eur.*, Mar. 2008, pp. 1069–1074.
- [60] R. S. Chakraborty and S. Bhunia, "HARPOON: An obfuscation-based SoC design methodology for hardware protection," *IEEE Trans. Comput.-Aided Design Integr. Circuits Syst.*, vol. 28, no. 10, pp. 1493–1502, Oct. 2009.
- [61] F. Koushanfar, S. Fazzari, C. McCants, W. Bryson, P. Song, M. Sale, and M. Potkonjak, "Can EDA combat the rise of electronic counterfeiting?" in *Proc. 49th Annu. Design Autom. Conf.*, Jun. 2012, pp. 133–138.
- [62] P. Kocher, J. Jaffe, and B. Jun, "Differential power analysis," in *Advances in Cryptology—CRYPTO'99: 19th Annual International Cryptology Conference Santa Barbara, California, USA, August 15–19, 1999 Proceedings 19*. Berlin, Germany: Springer, 1999, pp. 388–397.
- [63] P. Rohatgi, "Electromagnetic attacks and countermeasures," in *Cryptographic Engineering*, Ç. K. Koç, Ed. Springer, 2009, pp. 407–430. [Online]. Available: <https://link.springer.com/book/10.1007/978-0-387-71817-0>
- [64] A. Schlösser, D. Nedospasov, J. Krämer, S. Orlic, and J.-P. Seifert, "Simple photonic emission analysis of AES," *J. Cryptograph. Eng.*, vol. 3, pp. 3–15, 2013.
- [65] D. Genkin, A. Shamir, and E. Tromer, "RSA key extraction via low-bandwidth acoustic cryptanalysis," in *Advances in Cryptology—CRYPTO 2014: 34th Annual Cryptology Conference, Santa Barbara, CA, USA, August 17–21, 2014, Proceedings, Part I 34*. Berlin, Germany: Springer, 2014, pp. 444–461.
- [66] E. Brier, C. Clavier, and F. Olivier, "Correlation power analysis with a leakage model," in *Cryptographic Hardware and Embedded Systems—CHES 2004: 6th International Workshop Cambridge, MA, USA, August 11–13, 2004. Proceedings 6*. Berlin, Germany: Springer, 2004, pp. 16–29.
- [67] E. Peeters, F.-X. Standaert, and J.-J. Quisquater, "Power and electro-magnetic analysis: Improved model, consequences and comparisons," *Integration*, vol. 40, no. 1, pp. 52–60, Jan. 2007.
- [68] D. Blair, J. Huntsman, C. Barrett, W. Lynn, S. Gorton, D. Wince-Smith, and M. Young, "Update to the IP commission report," Commission Theft Amer. Intellectual Property, Nat. Bur. Asian Res., 2017.
- [69] P. Mall, R. Amin, A. K. Das, M. T. Leung, and K. R. Choo, "PUF-based authentication and key agreement protocols for IoT, WSNs, and smart grids: A comprehensive survey," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8205–8228, Jun. 2022.
- [70] B. Dey, K. Khalil, A. Kumar, and M. Bayoumi, "A novel design gate based low-cost configurable RO PUF using reversible logic," in *Proc. IEEE 62nd Int. Midwest Symp. Circuits Syst. (MWSCAS)*, Aug. 2019, pp. 211–214.
- [71] K. Shamsi, M. Li, K. Plaks, S. Fazzari, D. Z. Pan, and Y. Jin, "IP protection and supply chain security through logic obfuscation: A systematic overview," *ACM Trans. Des. Autom. Electron. Syst.*, vol. 24, no. 6, pp. 1–36, Nov. 2019.
- [72] S. Ellicott, M. Kines, W. Khalil, Y. Qi, A. Kurtoglu, and H. M. Lavasani, "Analog-inspired hardware security: A low-energy solution for IoT trusted communications," in *Proc. IEEE 34th Int. Syst.-on-Chip Conf. (SOCC)*, Sep. 2021, pp. 200–205.
- [73] X. Fan, "Wafer level packaging (WLP): Fan-in, fan-out and three-dimensional integration," in *Proc. 11th Int. Thermal, Mech. Multi-Phys. Simulation, Exp. Microelectron. Microsyst. (EuroSimE)*, Apr. 2010, pp. 1–7.
- [74] M. Bohr, "The new era of scaling in an SoC world," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2009, pp. 23–28.
- [75] C. Xi, N. Jessurun, and N. Asadizanjani, "A framework to assess the security of advanced integrated circuit (IC) packaging," in *Proc. IEEE 8th Electron. Syst.-Integr. Technol. Conf. (ESTC)*, Sep. 2020, pp. 1–7.
- [76] N. Vashistha, M. T. Rahman, O. P. Dizon-Paradis, and N. Asadizanjani, "Is backside the new backdoor in modern SoCs? Invited paper," in *Proc. IEEE Int. Test Conf. (ITC)*, Nov. 2019, pp. 1–10.
- [77] H. Kim, C. Hahn, and J. Hur, "Real-time detection of cache side-channel attack using non-cache hardware events," in *Proc. Int. Conf. Inf. Netw. (ICOIN)*, Jan. 2021, pp. 28–31.
- [78] B. Xu, W. Wang, Q. Hao, Z. Zhang, P. Du, T. Xia, H. Li, and X. Wang, "A security design for the detecting of buffer overflow attacks in IoT device," *IEEE Access*, vol. 6, pp. 72862–72869, 2018.
- [79] Z. Hongwei, K. Zhipeng, Z. Yuchen, G. Ruichao, L. Shengsheng, and Y. Jinhui, "RS-stack: Defense against stack buffer overflow attack with random shadow stack," in *Proc. IEEE 2nd Int. Conf. Data Sci. Comput. Appl. (ICDSCA)*, Oct. 2022, pp. 1432–1436.
- [80] T. Han, Y. Wang, and P. Liu, "Hardware trojans detection at register transfer level based on machine learning," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2019, pp. 1–5.
- [81] S. Ainsworth and T. M. Jones, "MuonTrap: Preventing cross-domain spectre-like attacks by capturing speculative state," in *Proc. ACM/IEEE 47th Annu. Int. Symp. Comput. Archit. (ISCA)*, May 2020, pp. 132–144.
- [82] V. J. Kulkarni, R. Manju, R. Gupta, J. Jose, and S. Nandi, "Packet header attack by hardware trojan in NoC based TCMP and its impact analysis," in *Proc. 15th IEEE/ACM Int. Symp. Netw.-on-Chip (NOCS)*, Oct. 2021, pp. 21–28.
- [83] A. Jain and U. Guin, "A novel tampering attack on AES cores with hardware trojans," in *Proc. IEEE Int. Test Conf. Asia (ITC-Asia)*, Sep. 2020, pp. 77–82.
- [84] M. Elshamy, G. Di Natale, A. Pavlidis, M. Louërat, and H.-G. Stratigopoulos, "Hardware trojan attacks in analog/mixed-signal ICs via the test access mechanism," in *Proc. IEEE Eur. Test Symp. (ETS)*, May 2020, pp. 1–6.
- [85] M. A. Khelif, J. Lorandel, O. Romain, M. Regnery, D. Baheux, and G. Barbu, "Toward a hardware man-in-the-middle attack on PCIe bus for smart data replay," in *Proc. 22nd Euromicro Conf. Digit. Syst. Design (DSD)*, Aug. 2019, pp. 230–237.
- [86] M.-H. Kuo, C.-M. Hu, and K.-J. Lee, "Time-related hardware trojan attacks on processor cores," in *Proc. IEEE Int. Test Conf. Asia (ITC-Asia)*, Sep. 2019, pp. 43–48.

- [87] M. Gay, T. Paxian, D. Upadhyaya, B. Becker, and I. Polian, "Hardware-oriented algebraic fault attack framework with multiple fault injection support," in *Proc. Workshop Fault Diagnosis Tolerance Cryptogr. (FDTC)*, Aug. 2019, pp. 25–32.
- [88] X. Chen, C. Liu, B. Li, K. Lu, and D. Song, "Targeted backdoor attacks on deep learning systems using data poisoning," 2017, *arXiv:1712.05526*.
- [89] Y. Liu, S. Ma, Y. Aafer, W.-C. Lee, J. Zhai, W. Wang, and X. Zhang, "Trojaning attack on neural networks," in *Proc. 25th Annu. Netw. Distrib. Syst. Secur. Symp.*, 2017, pp. 1–16.
- [90] Y. Kim, R. Daly, J. Kim, C. Fallin, J. H. Lee, D. Lee, C. Wilkerson, K. Lai, and O. Mutlu, "Flipping bits in memory without accessing them: An experimental study of DRAM disturbance errors," *ACM SIGARCH Comput. Archit. News*, vol. 42, no. 3, pp. 361–372, Jun. 2014.
- [91] K. Cai, M. H. I. Chowdhury, Z. Zhang, and F. Yao, "Seeds of SEED: NMT-stroke: Diverting neural machine translation through hardware-based faults," in *Proc. Int. Symp. Secure Private Execution Environ. Design (SEED)*, Sep. 2021, pp. 76–82.
- [92] K. Cai, Z. Zhang, and F. Yao, "On the feasibility of training-time trojan attacks through hardware-based faults in memory," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, Jun. 2022, pp. 133–136.
- [93] R. Mukherjee and R. S. Chakraborty, "Novel hardware trojan attack on activation parameters of FPGA-based DNN accelerators," *IEEE Embedded Syst. Lett.*, vol. 14, no. 3, pp. 131–134, Sep. 2022.
- [94] T. A. Odetola and S. R. Hasan, "SoWaF: Shuffling of weights and feature maps: A novel hardware intrinsic attack (HIA) on convolutional neural network (CNN)," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, May 2021, pp. 1–5.
- [95] B. Hettwer, D. Fennes, S. Leger, J. Richter-Brockmann, S. Gehrler, and T. Güneysu, "Deep learning multi-channel fusion attack against side-channel protected hardware," in *Proc. 57th ACM/IEEE Design Autom. Conf. (DAC)*, Jul. 2020, pp. 1–6.
- [96] N. Ivanov and Q. Yan, "EthClipper: A clipboard meddling attack on hardware wallets with address verification evasion," in *Proc. IEEE Conf. Commun. Netw. Secur. (CNS)*, Oct. 2021, pp. 191–199.
- [97] F. Almeida, M. Imran, J. Raik, and S. Pagliarini, "Ransomware attack as hardware trojan: A feasibility and demonstration study," *IEEE Access*, vol. 10, pp. 44827–44839, 2022.
- [98] Q. Chen and R. A. Bridges, "Automated behavioral analysis of malware: A case study of WannaCry ransomware," in *Proc. 16th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA)*, Dec. 2017, pp. 454–460.
- [99] U. Ali and O. Khan, "ConNOC: A practical timing channel attack on network-on-chip hardware in a multicore processor," in *Proc. IEEE Int. Symp. Hardw. Oriented Secur. Trust (HOST)*, Dec. 2021, pp. 192–202.

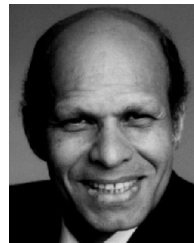


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